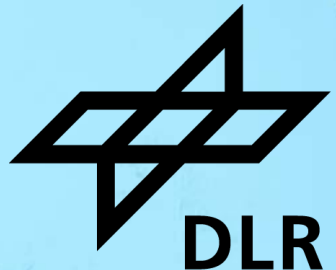


INTRODUCTION TO DEEP LEARNING

PART II – DEEP GENERATIVE MODELS

Auliya Fitri, Sheetal Sahoo, Sreerag Naveenachandran

**Machine Learning Group
Institute of Data Science**



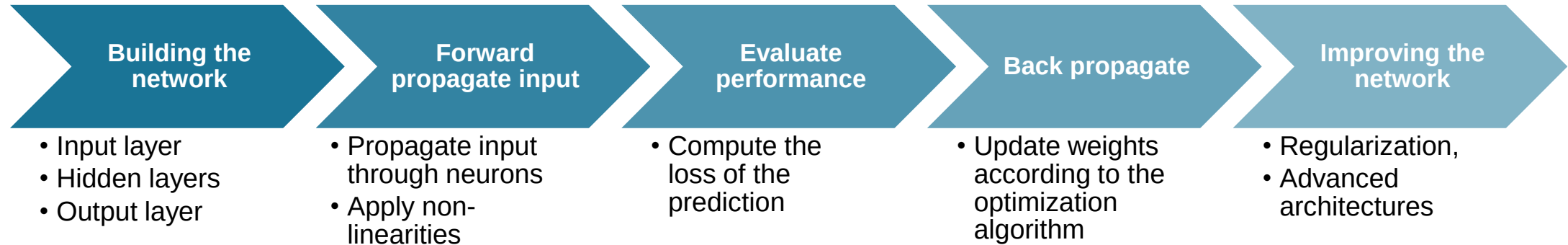
Schedule



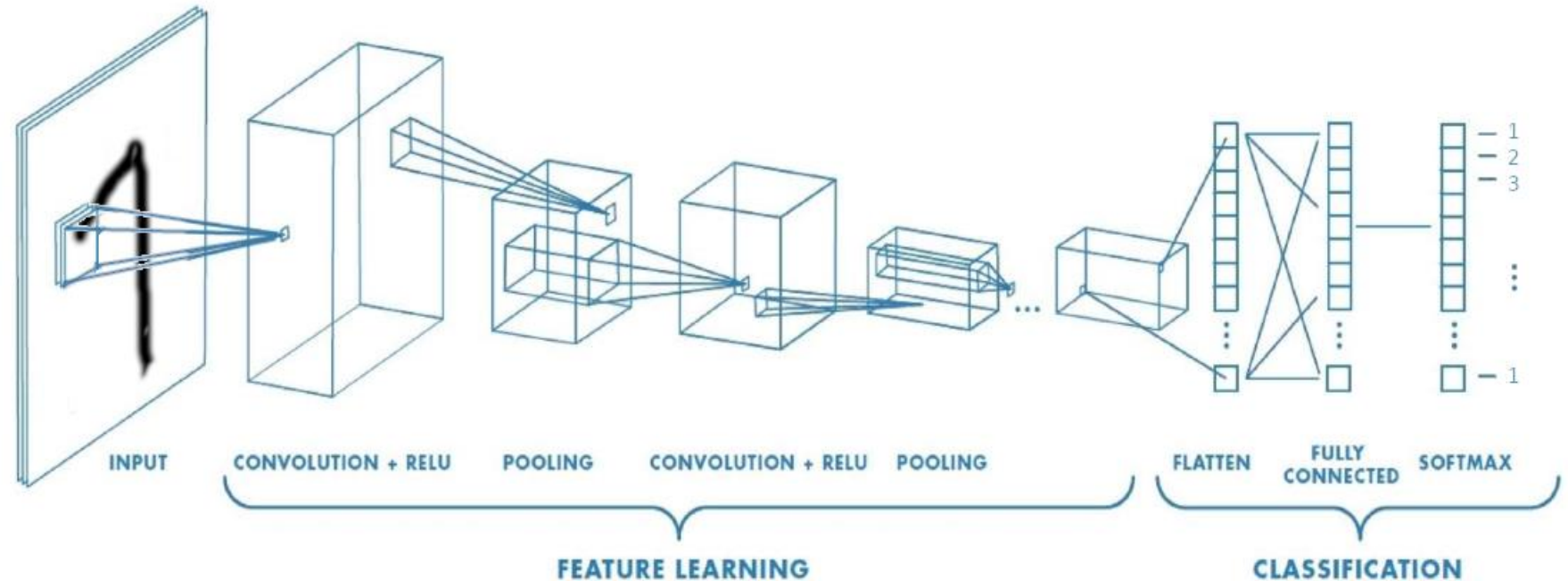
Date	Time	Activity
28.05.2026 Day 1	09:00 - 10:00	Introduction and basics
	10:00 - 10:30	Hands-on I
	10:30 - 10:45	Coffee break
	10:45 - 11:15	Hands-on II
	11:15 - 12:00	Hands-on III (Group Work)
	12:00 - 13:00	Lunch break
	13:00 - 14:00	Advanced concept and Convolutional Neural Network
	14:00 - 14:45	Hands-on IV
	14:45 - 15:15	Recap Day 1
29.05.2026 Day 2	09:00 - 10:00	Deep Generative Models
	10:00 - 10:45	Hands-on III
	10:45 - 11:15	Latest applications, code and knowledge sources
	11:15 - 11:30	Closing



Recap Day 1: Neural network concepts



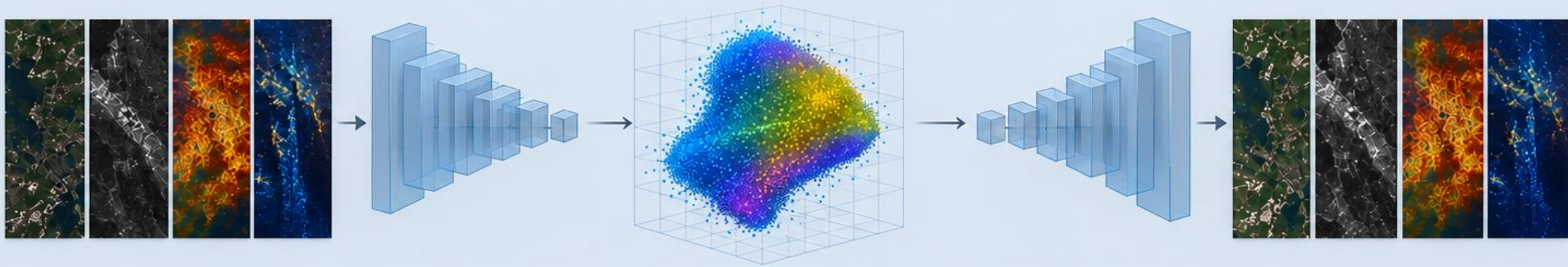
Convolutional Neural Network



- CONV and POOL layers output high-level features of input
- Fully connected layer uses these features for classifying input image
- Express output as **probability** of image belonging to a particular class

$$\text{softmax}(y_i) = \frac{e^{y_i}}{\sum_j e^{y_j}}$$

How can we learn the distribution of any kind of data to generate similar ones?



Curriculum



- Unsupervised learning
- Generative modeling
- Autoencoder (AE)
- Variational Autoencoder (VAE)

*some slides are adopted from MIT Deep learning course

Unsupervised learning

Unsupervised learning

How much information does the machine need to predict?

- Training samples depend on the amount of information predicted.
- More predictive information → larger and better learning models.
- Predicting only labels or rewards is not enough.
- Machines must learn representations of images, text, videos, and sensory data.

Aspect	Pure Reinforcement Learning (Cherry)	Supervised Learning (Icing)	Unsupervised /Predictive Learning (Cake)
What the machine predicts	A scalar reward given once in a while.	A category or a few numbers for each input.	Any part of its input for any observed part.
Data provided by	Environment (reward signal)	Human (labeled data)	The data itself
Information per sample	A few bits	10 → 10,000 bits per sample	Millions of bits per sample
Goal	Maximize cumulative reward	Learn mapping from input to target	Learn structure/patterns from data

“Predicting human-provided labels is not enough.”
 – Geoffrey Hinton

**Yann Lecunn's slides keynote NIPS 2016*

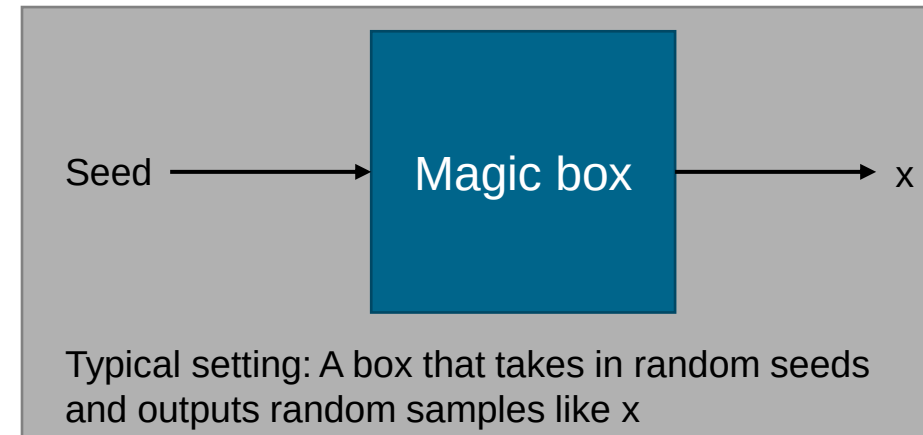
■ Supervised learning

- Data: (x,y)
- x is data, y is label
- Goal: Learn a function to map $x \rightarrow y$
- Examples: Classification, regression, object detection, semantic segmentation etc.

■ Unsupervised learning

- Data: x
- x is data, no labels!
- Goal: Learn the hidden or underlying structure of the data
- Examples: Clustering, feature or dimensionality reduction, generative modeling etc.

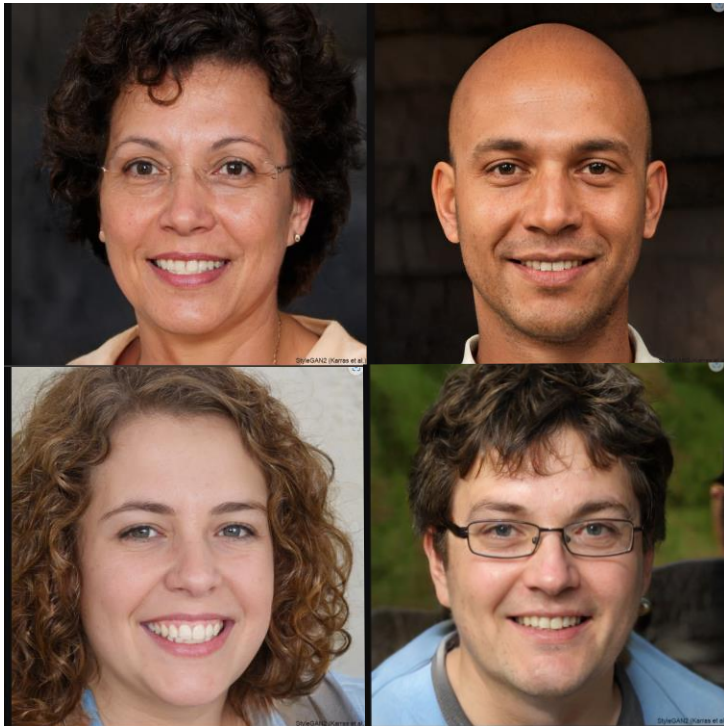
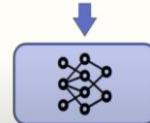
Generative modeling



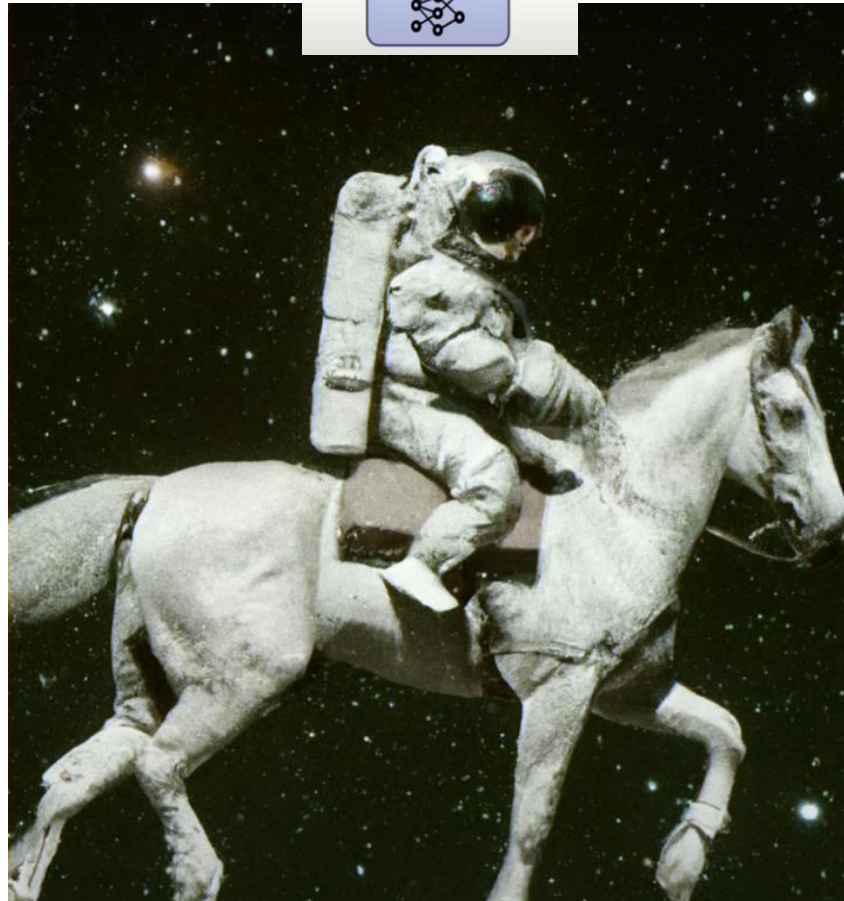
Generative modeling

Text-to-image / Text-to-Text / Image generation

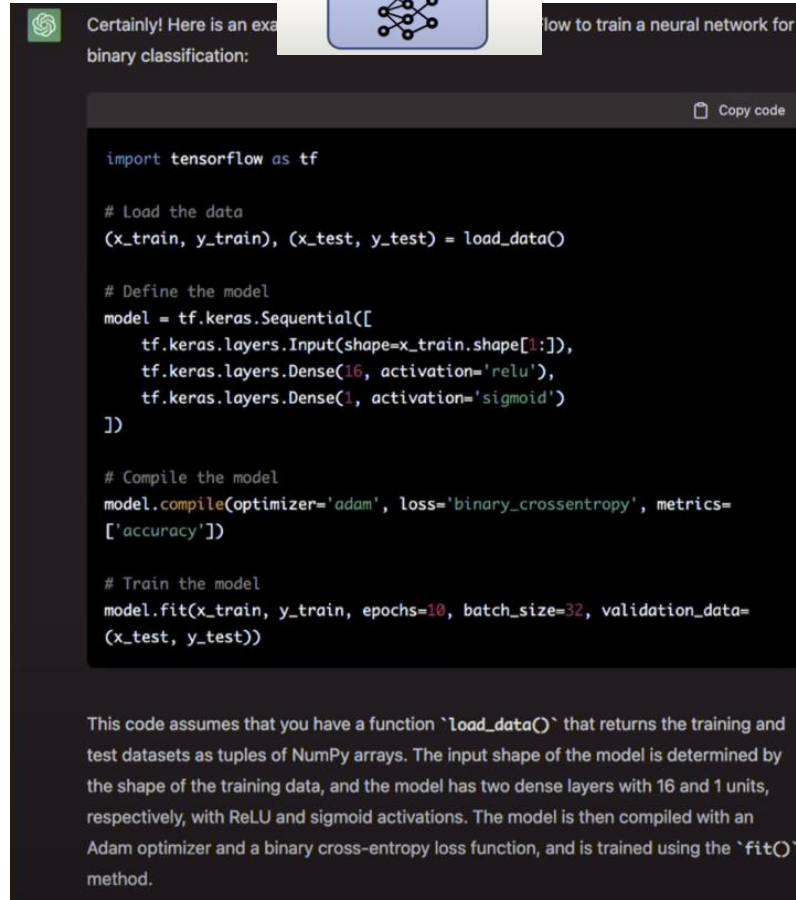
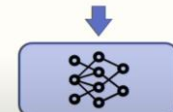
"A photo of an astronaut riding a horse."



*Goodfellow et.al, This person doesn't exist



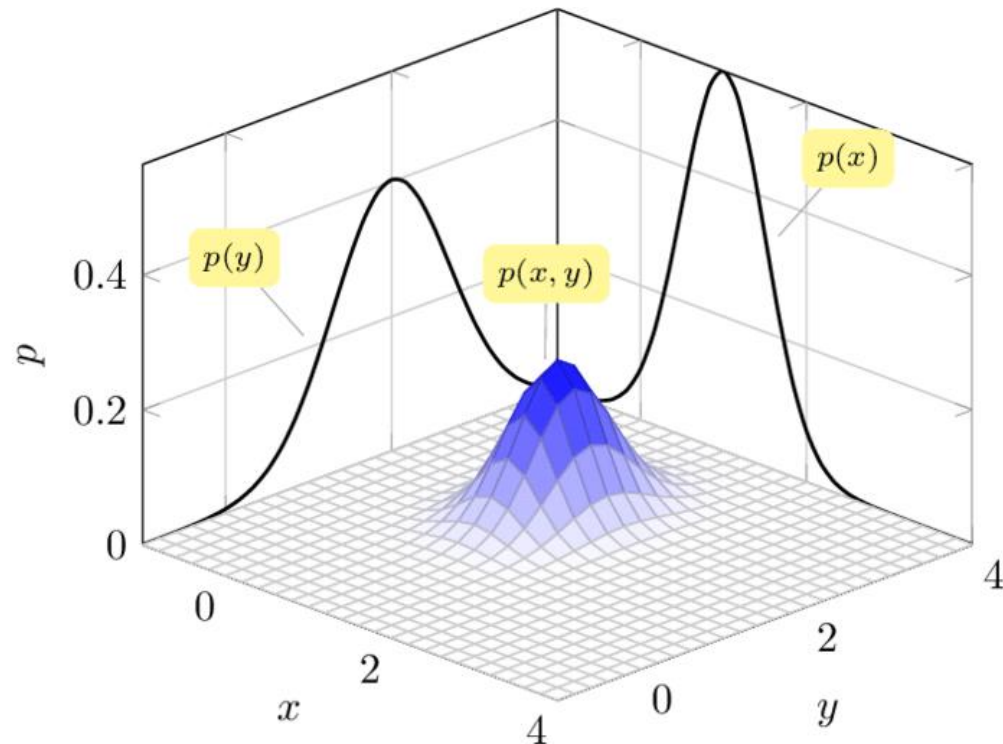
"Write code in TensorFlow to train a neural network."



Generative modeling

Data distribution

- Goal: Take as input training samples from some distribution and learn a model that represents that distribution



Input samples

Training data $\sim P_{data}(x)$



Generated samples

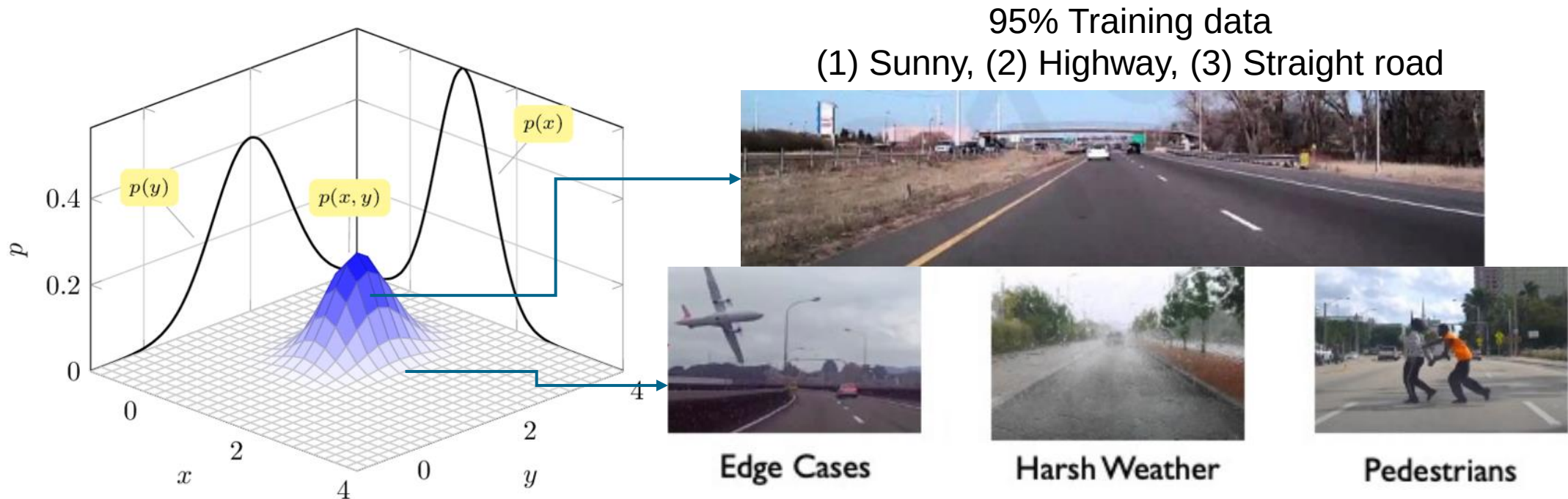
Generated $\sim P_{model}(x)$

How can we learn $P_{data}(x)$ similar to $P_{model}(x)$

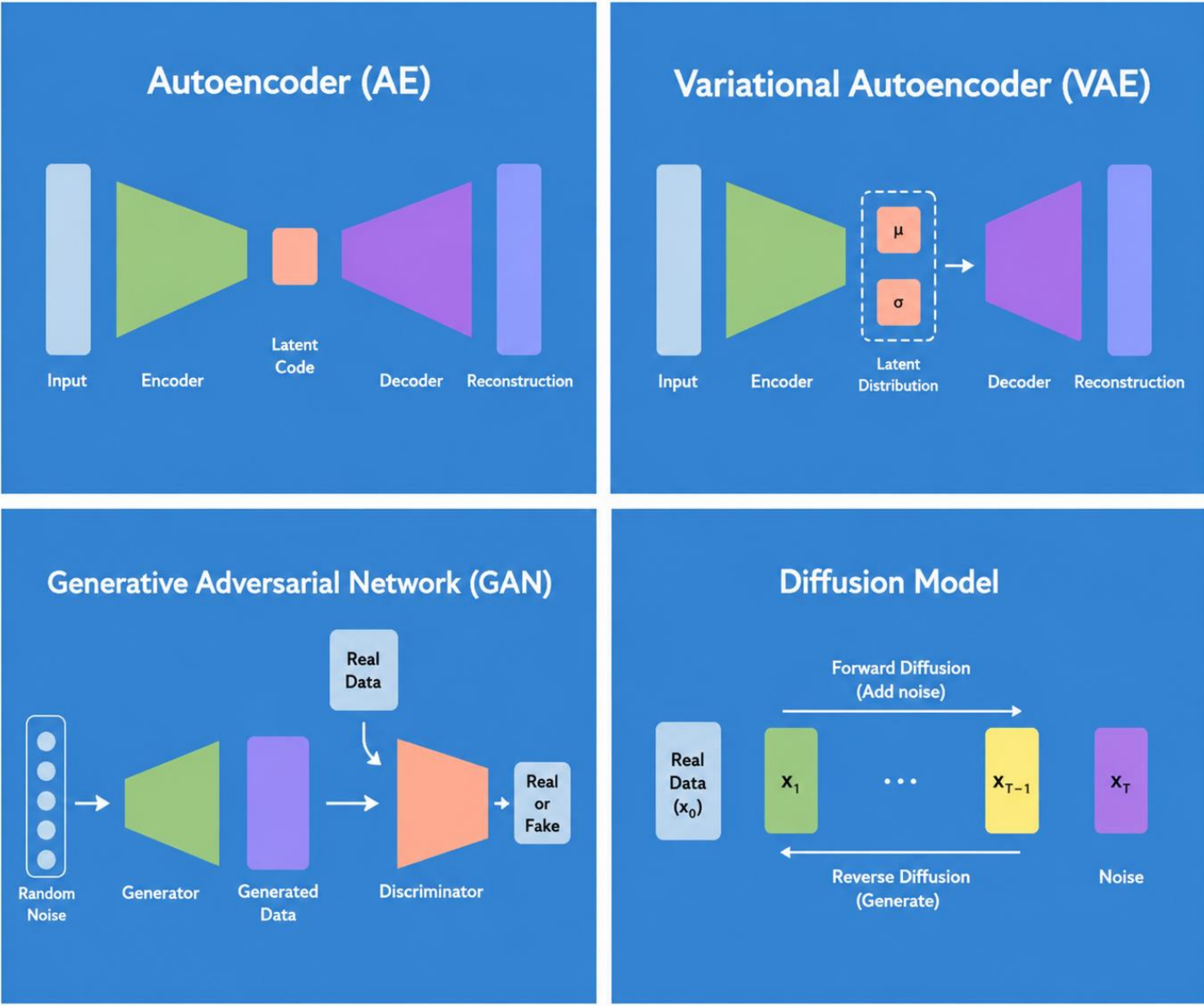
Generative modeling

Application: Outlier detection

- Problem: How can we detect when we encounter something new or rare?
- Strategy: Leverage generative models, detect outliers in the distribution
- Use outliers during training to improve even more



Generative models



Generative modeling

Latent variables



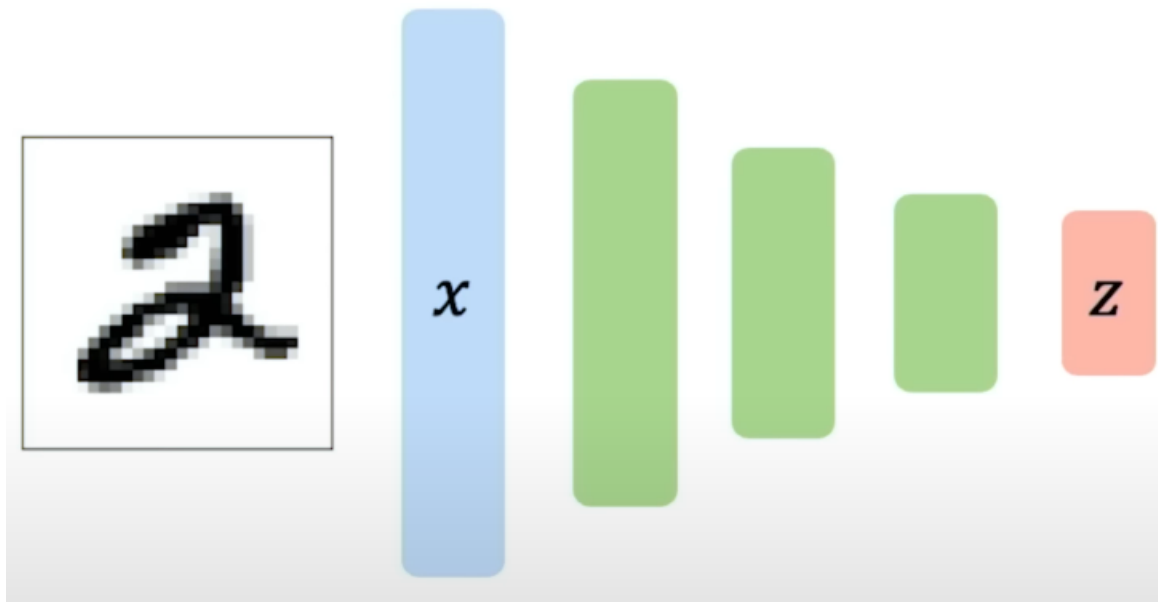
Can we learn the true explanatory factors, e.g. Latent variables, from only observed data?

Autoencoder

Autoencoder

Latent variables

- Unsupervised approach for learning a lower dimensional representation from unlabeled training data

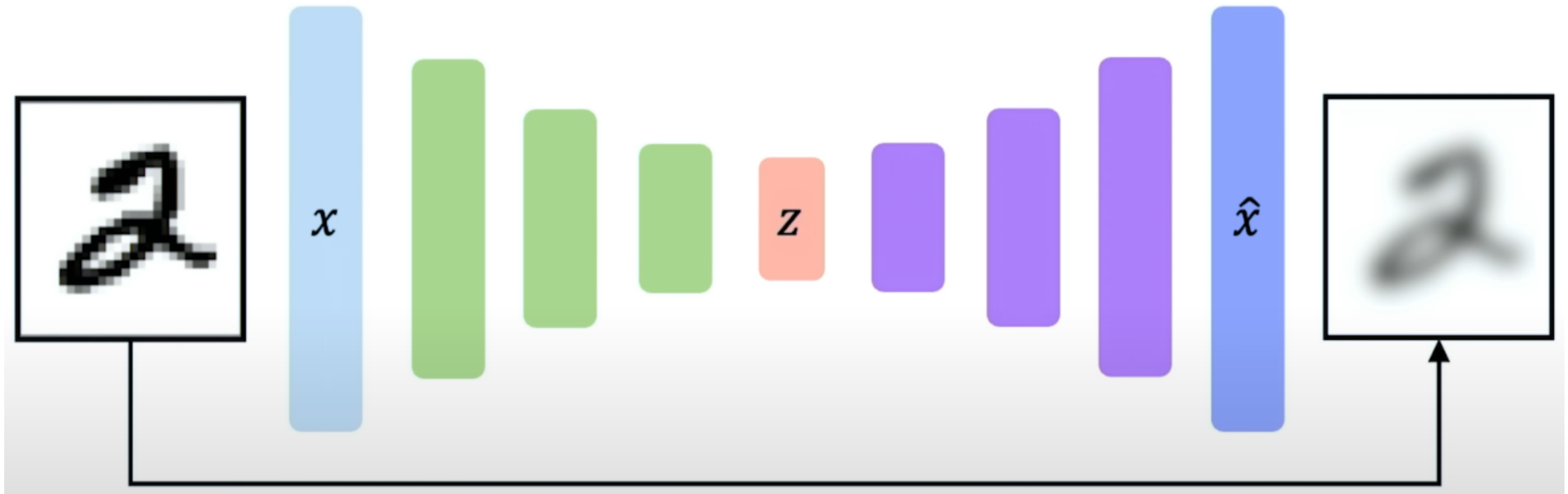


- “Encoder” learns mapping from the data, x , to a low dimensional latent space, z

Autoencoder

Data reconstruction

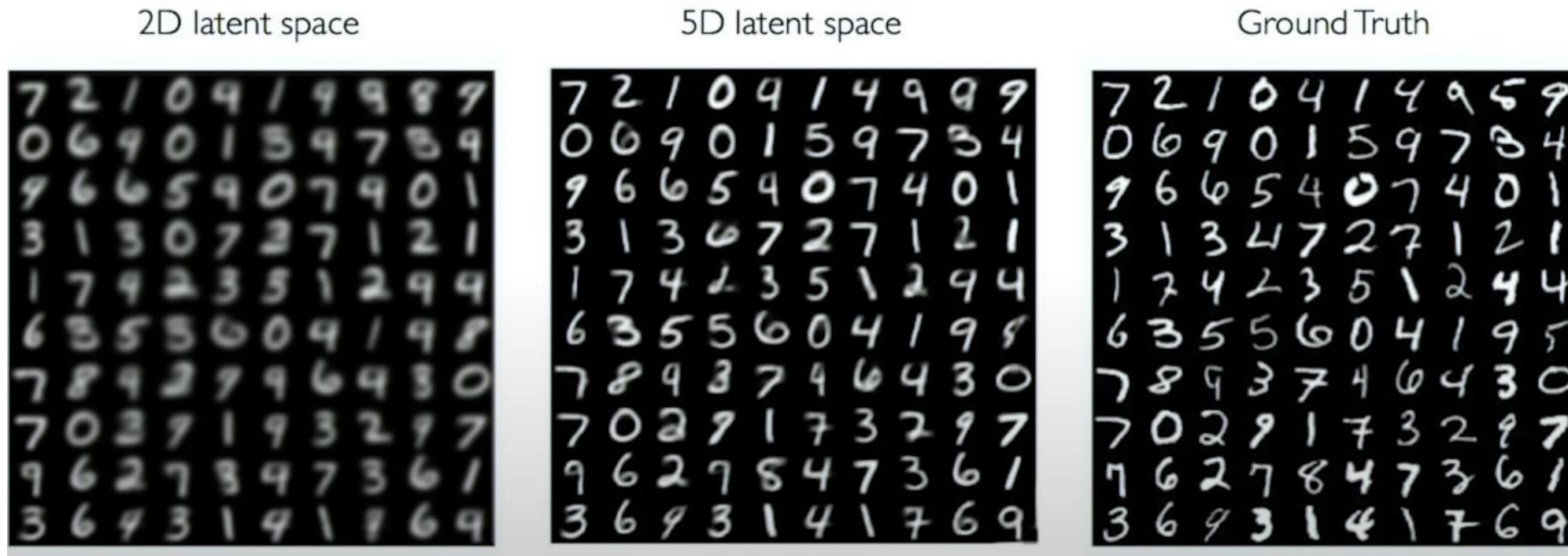
- How can we learn this latent space?
- Train the model to use these features to reconstruct the original data



$$\mathcal{L}(x, \hat{x}) = \|x - \hat{x}\|^2$$

Autoencoder Bottleneck

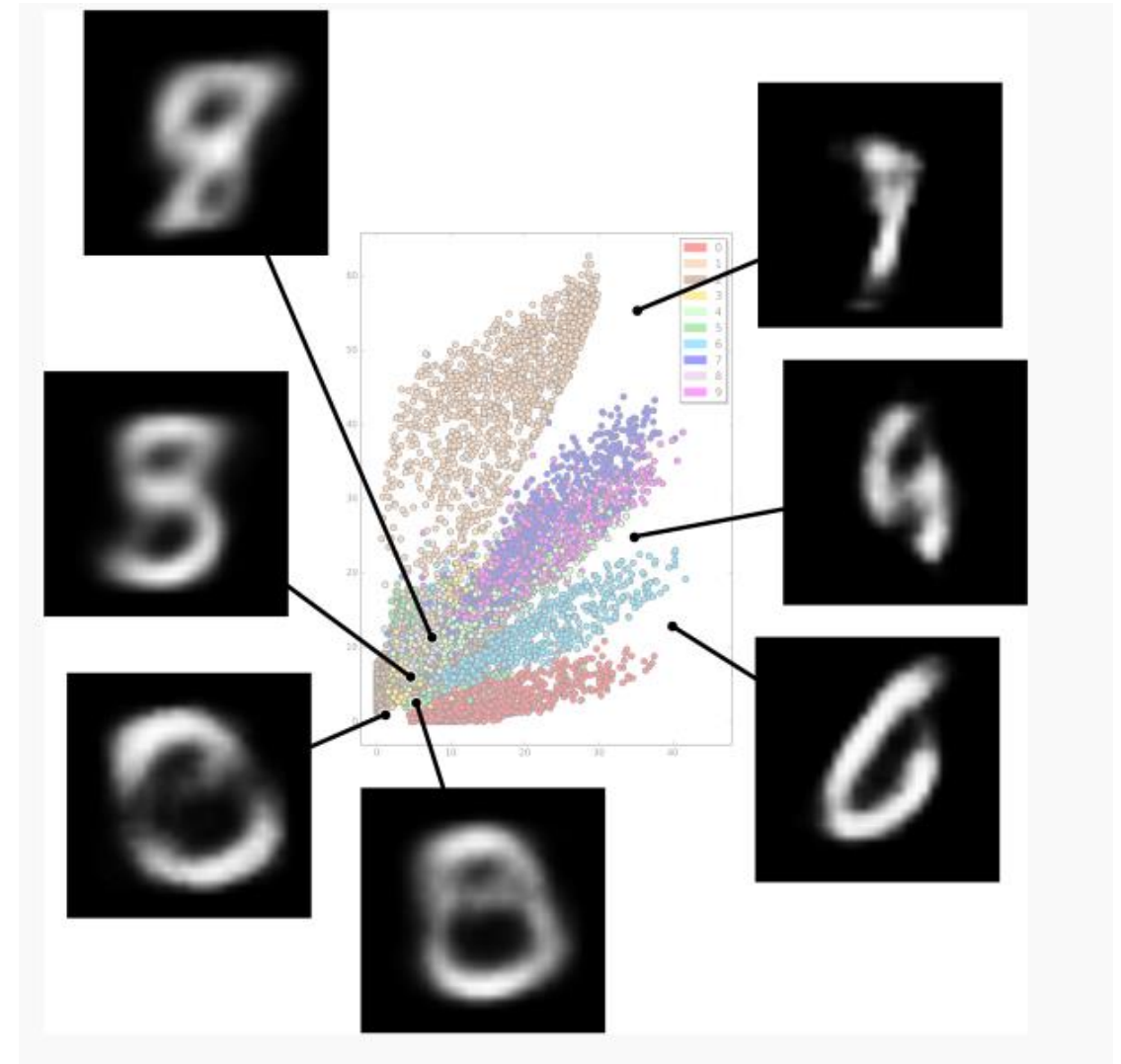
- Autoencoding is a form of compression! Smaller latent space will force a larger training bottleneck.



- Dimensionality of latent space → Reconstruction quality

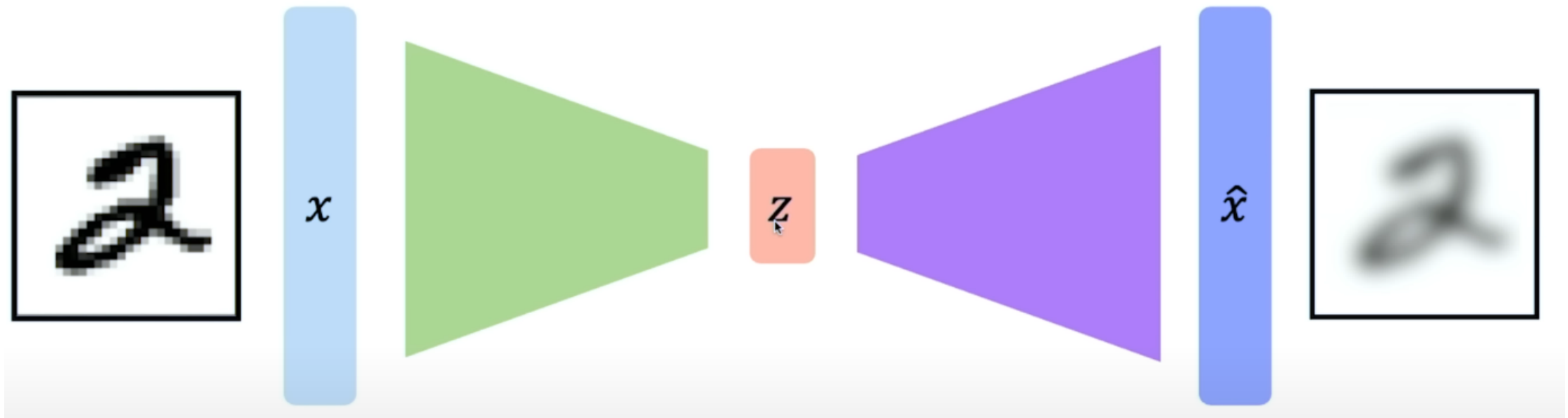
Autoencoder Bottleneck

- Bottleneck hidden layer forces network to learn a compressed latent representation
- Reconstruction loss forces the latent representation to capture (or encode) as much “information” about the data as possible
- Autoencoding = Automatically encode data; : “Auto” = Self-encoding



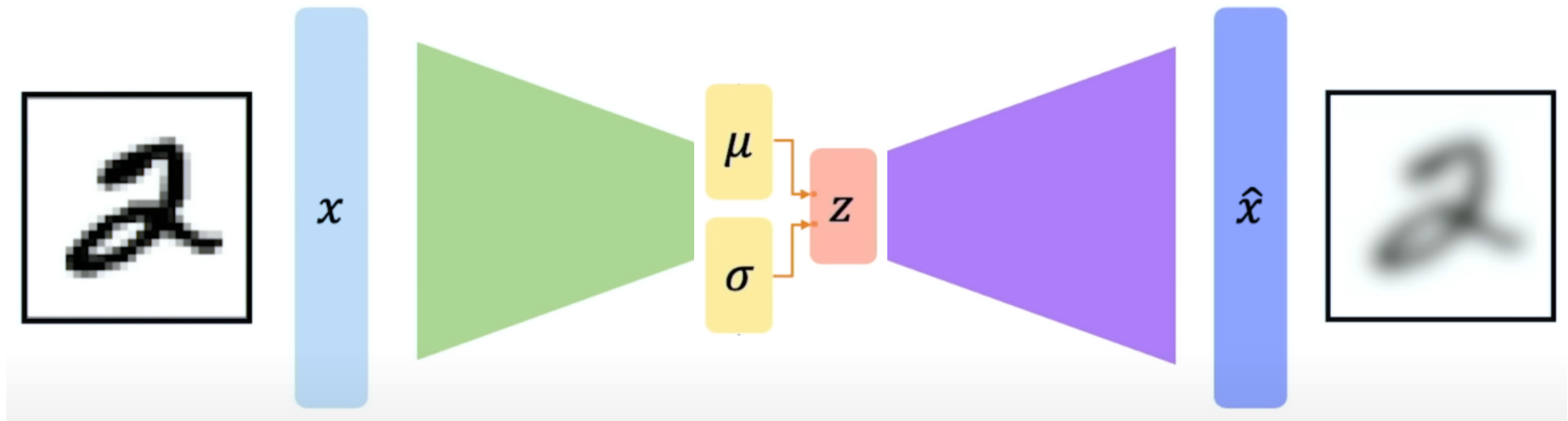
Variational Autoencoder (VAE)

Autoencoder



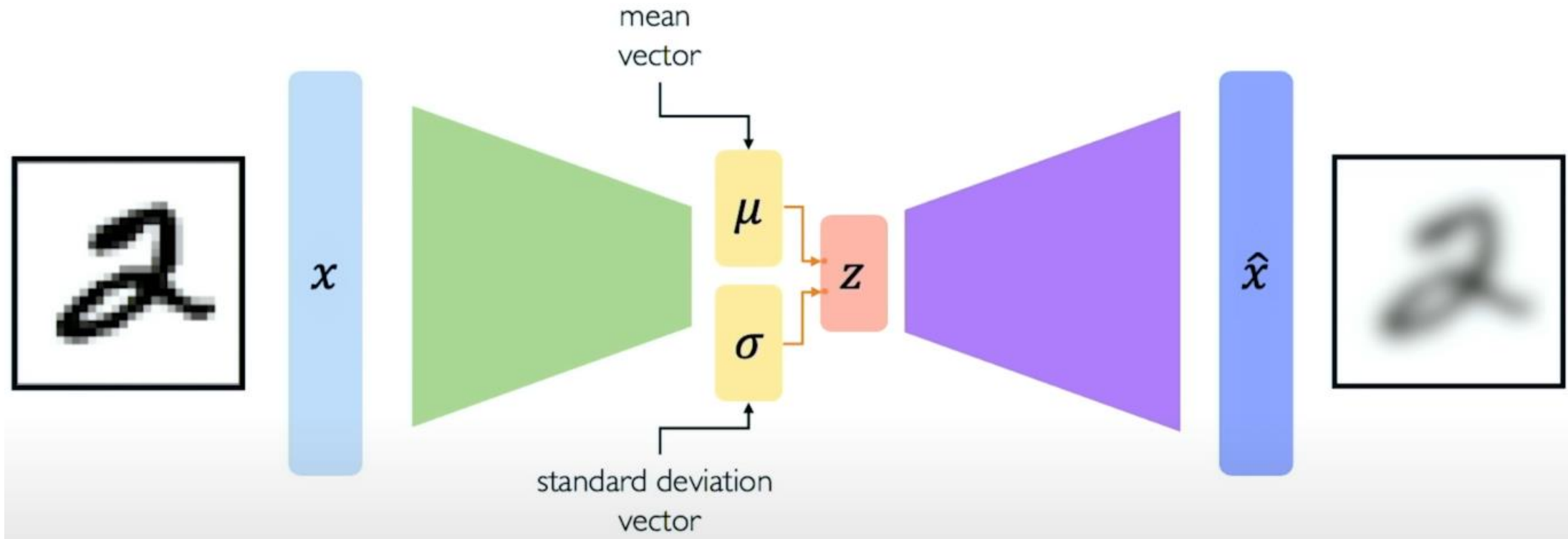
Autoencoder

Variational Autoencoder



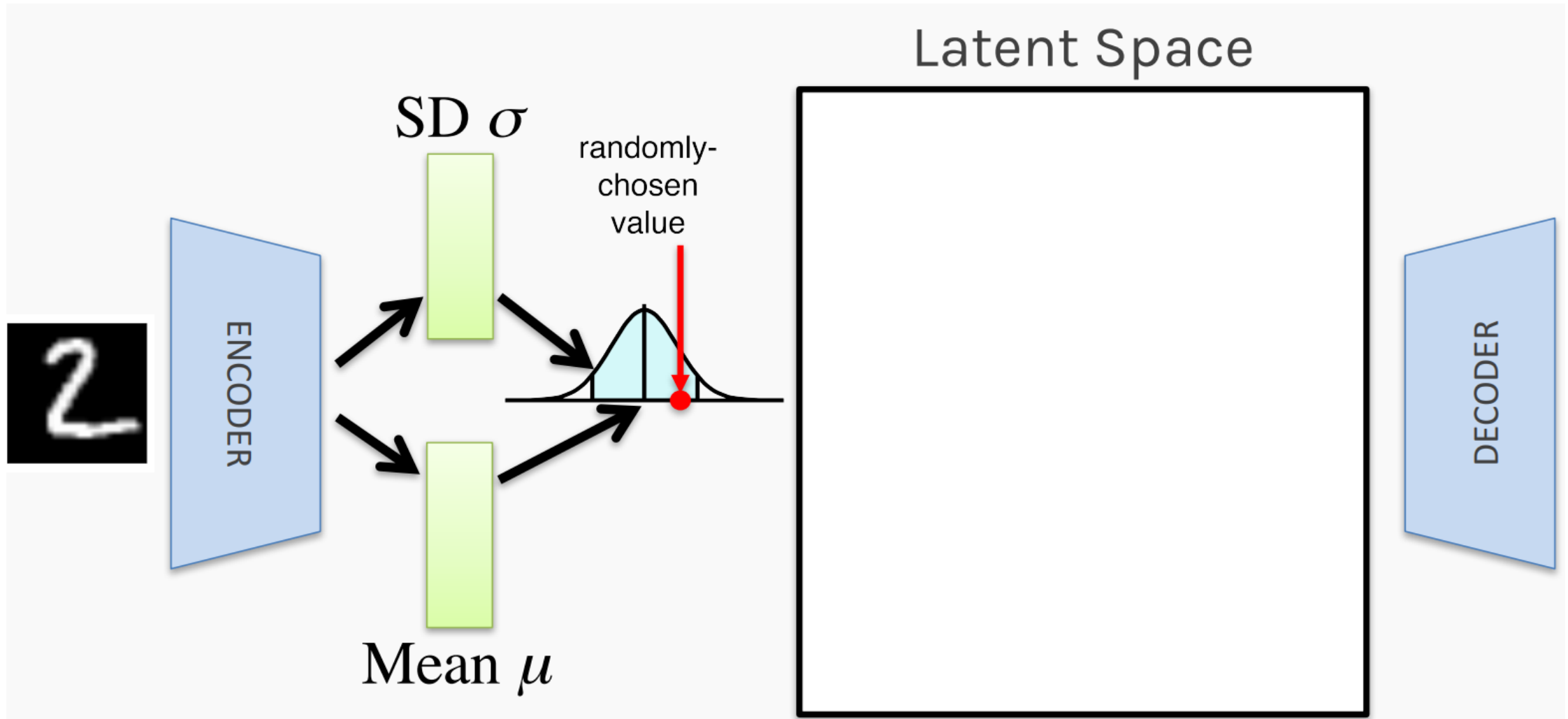
Variational Autoencoder

Variational Autoencoder

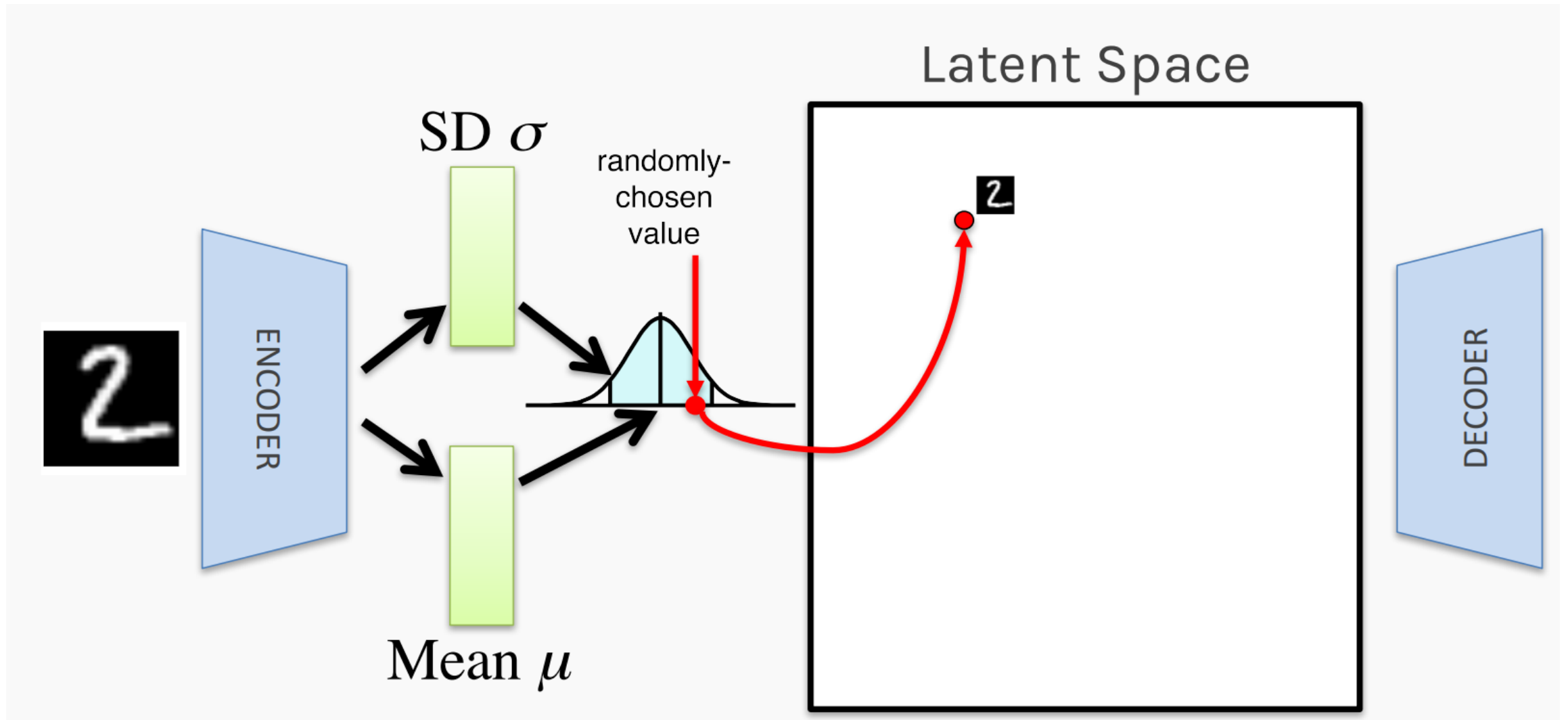


- Variational autoencoders are a probabilistic twist on autoencoders!
- Sample from the mean and standard deviation to compute latent space.

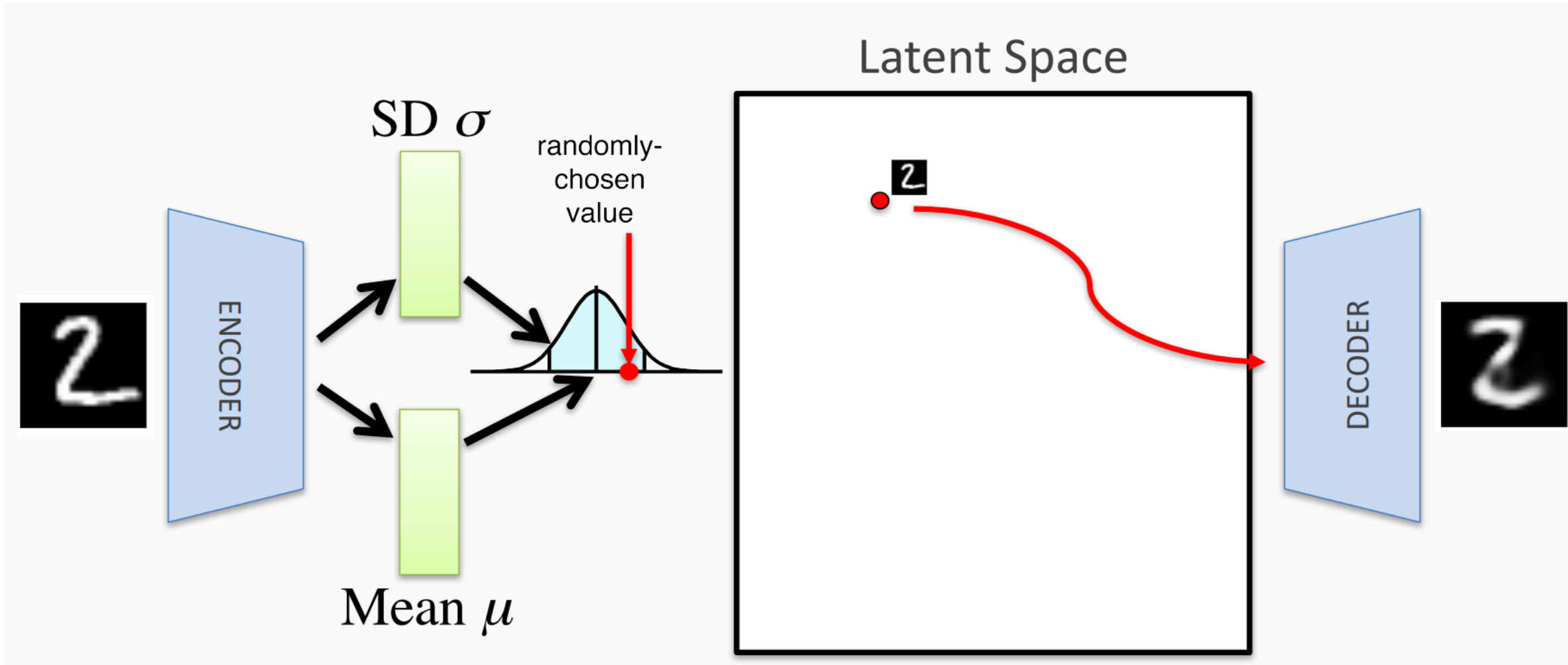
Variational Autoencoder



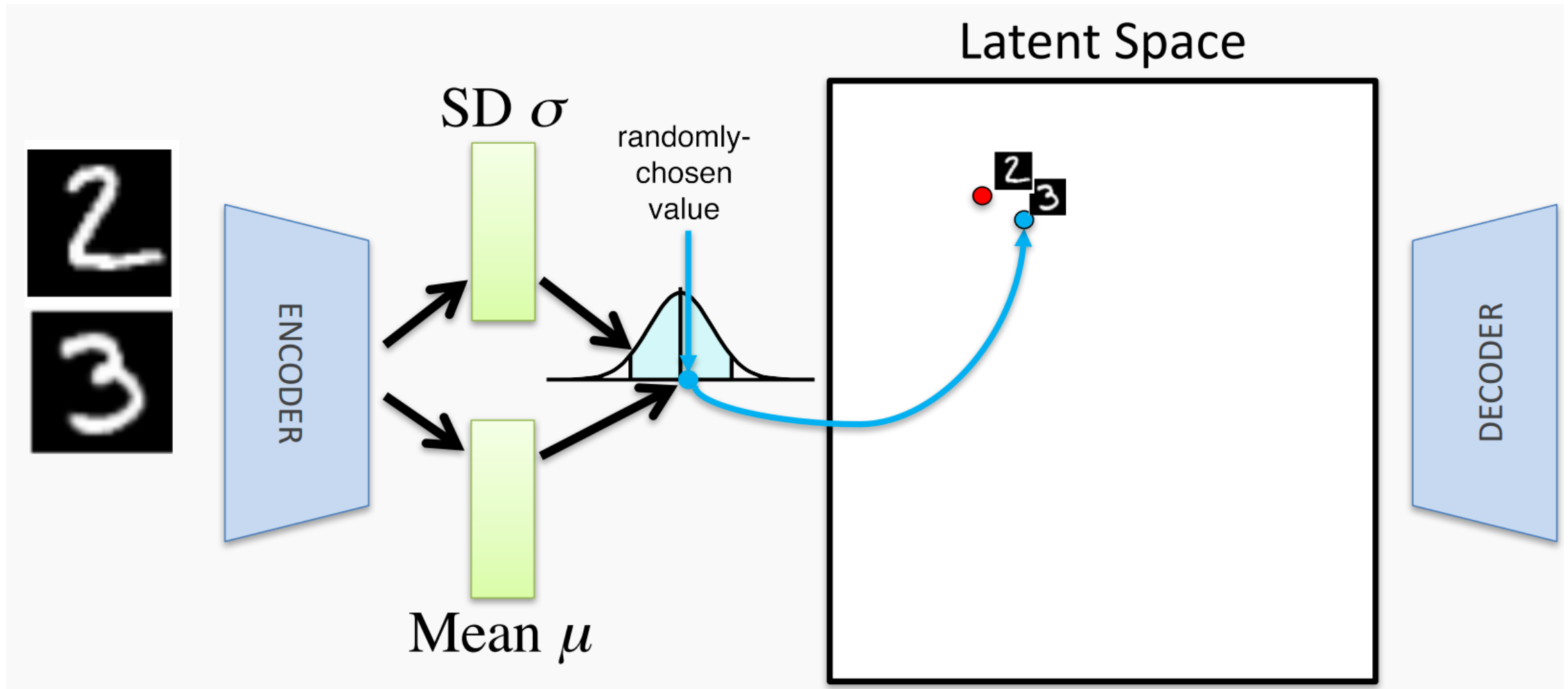
Variational Autoencoder



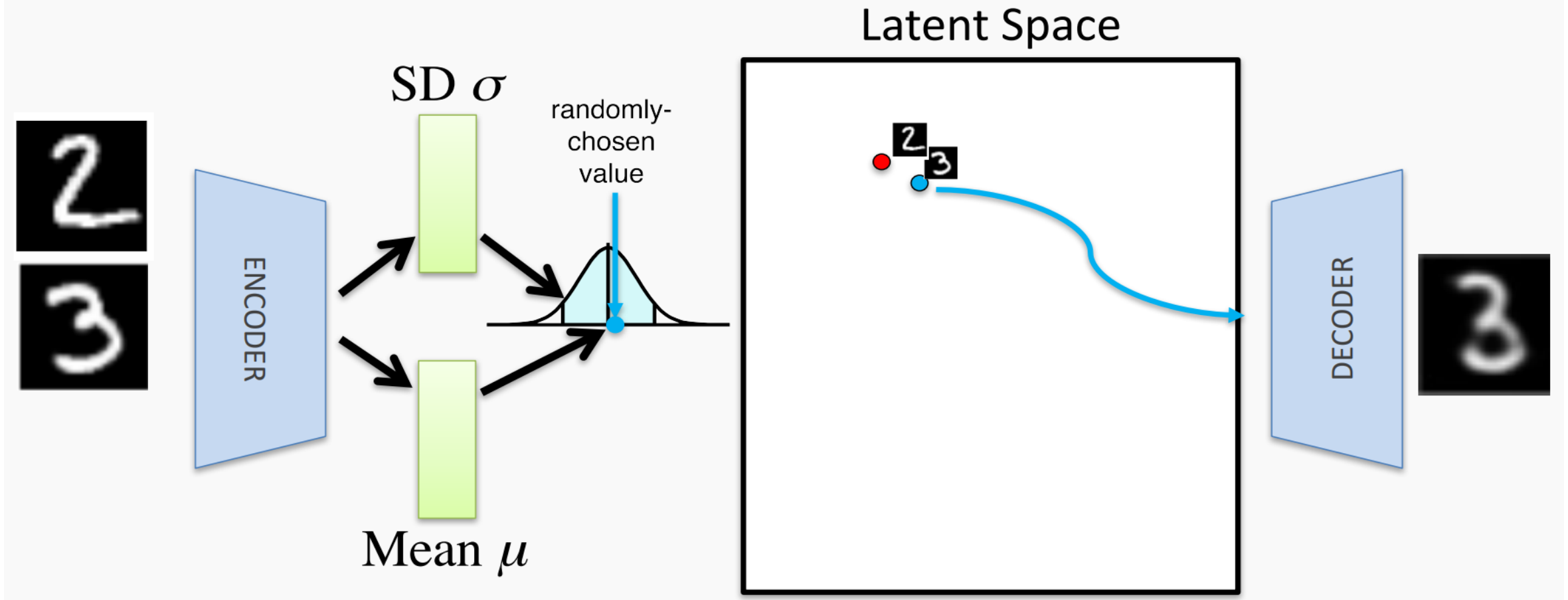
Variational Autoencoder



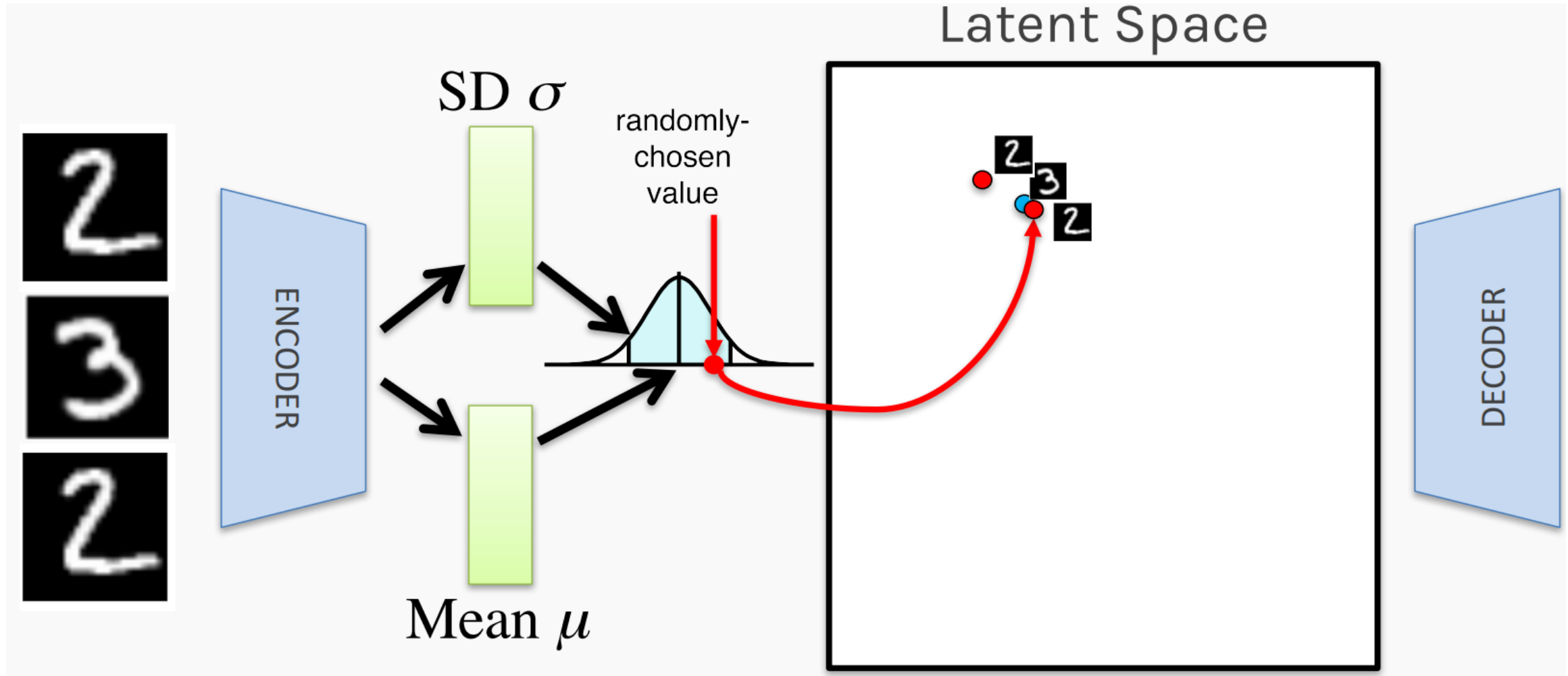
Variational Autoencoder



Variational Autoencoder



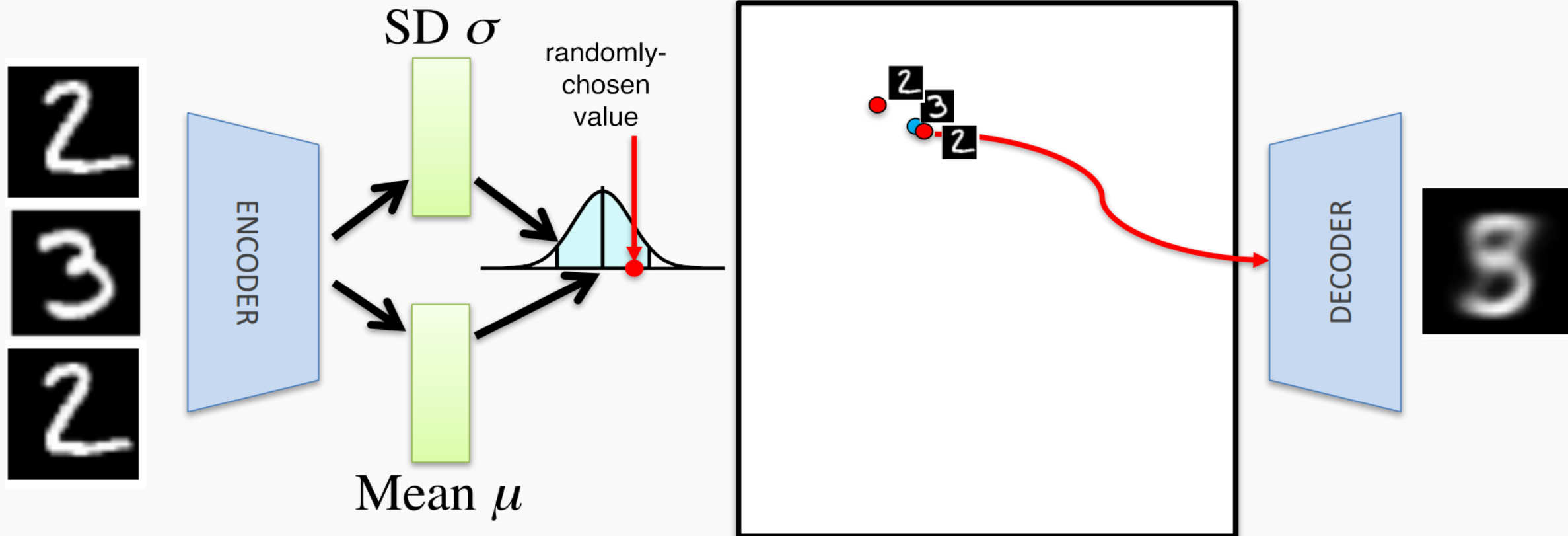
Variational Autoencoder



Variational Autoencoder

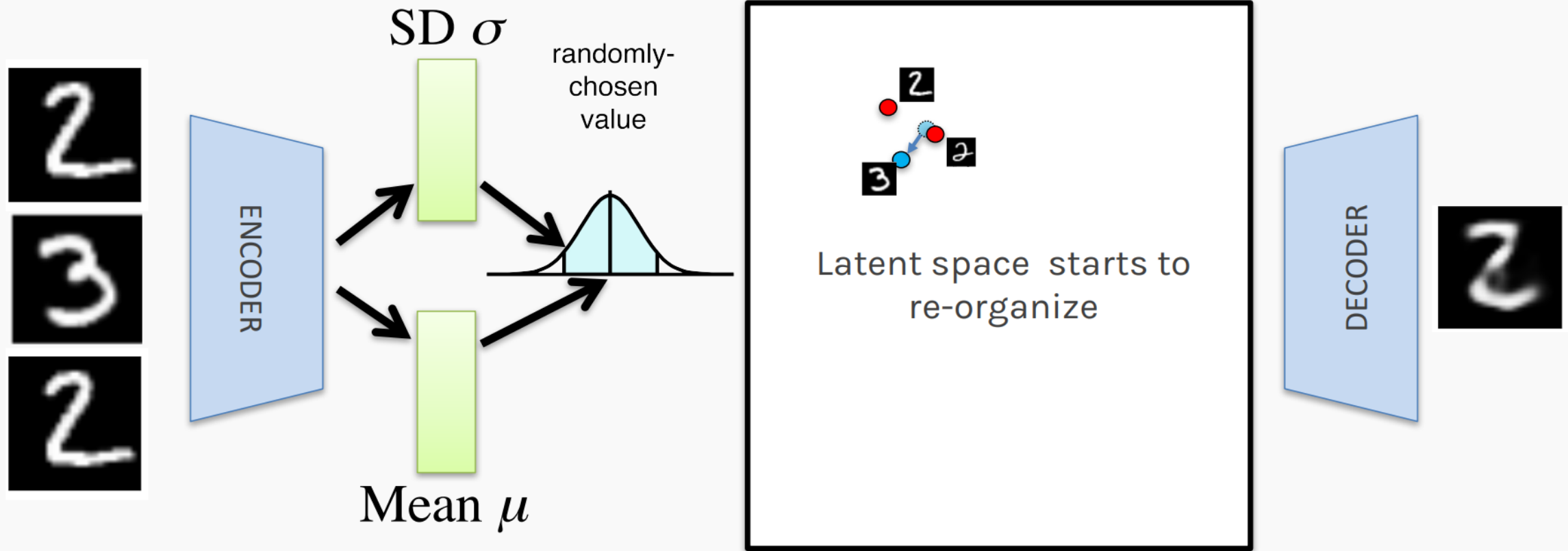
Train with 1st sample again

Latent Space



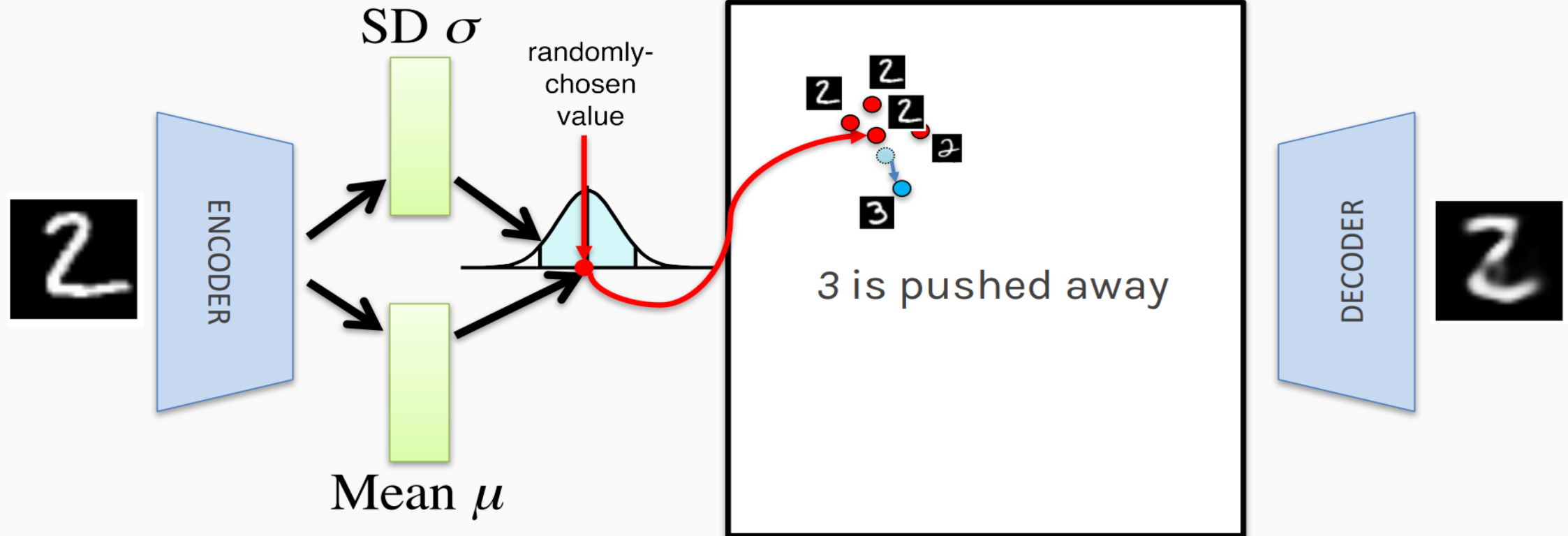
Variational Autoencoder

Train with 1st sample again



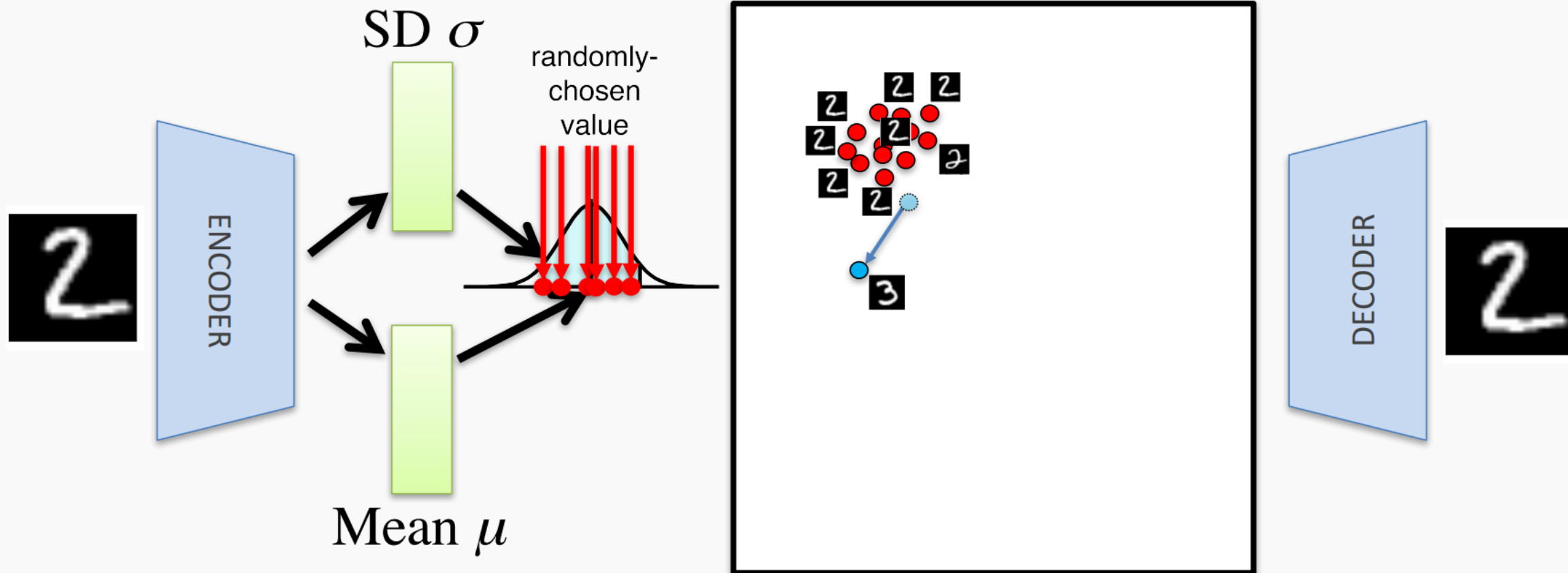
Variational Autoencoder

And again...



Variational Autoencoder

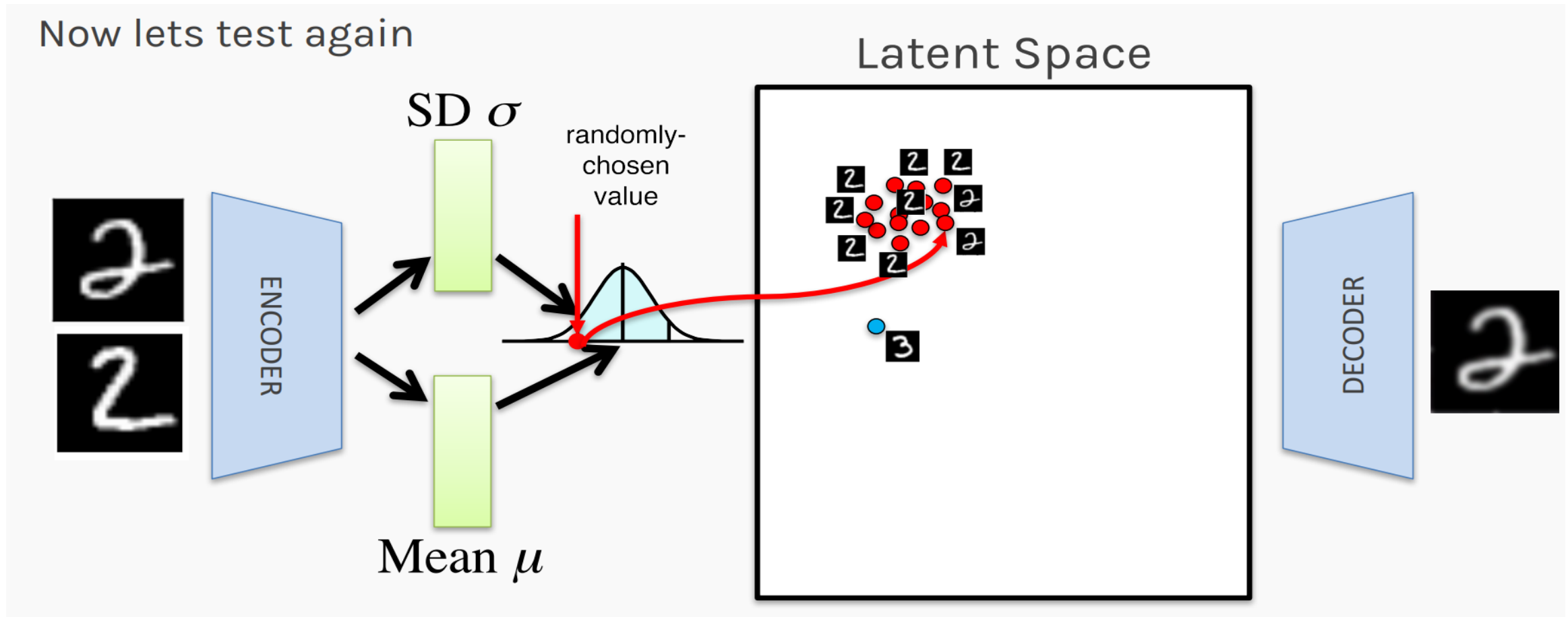
Many times...



Variational Autoencoder

- Blending the latent variables

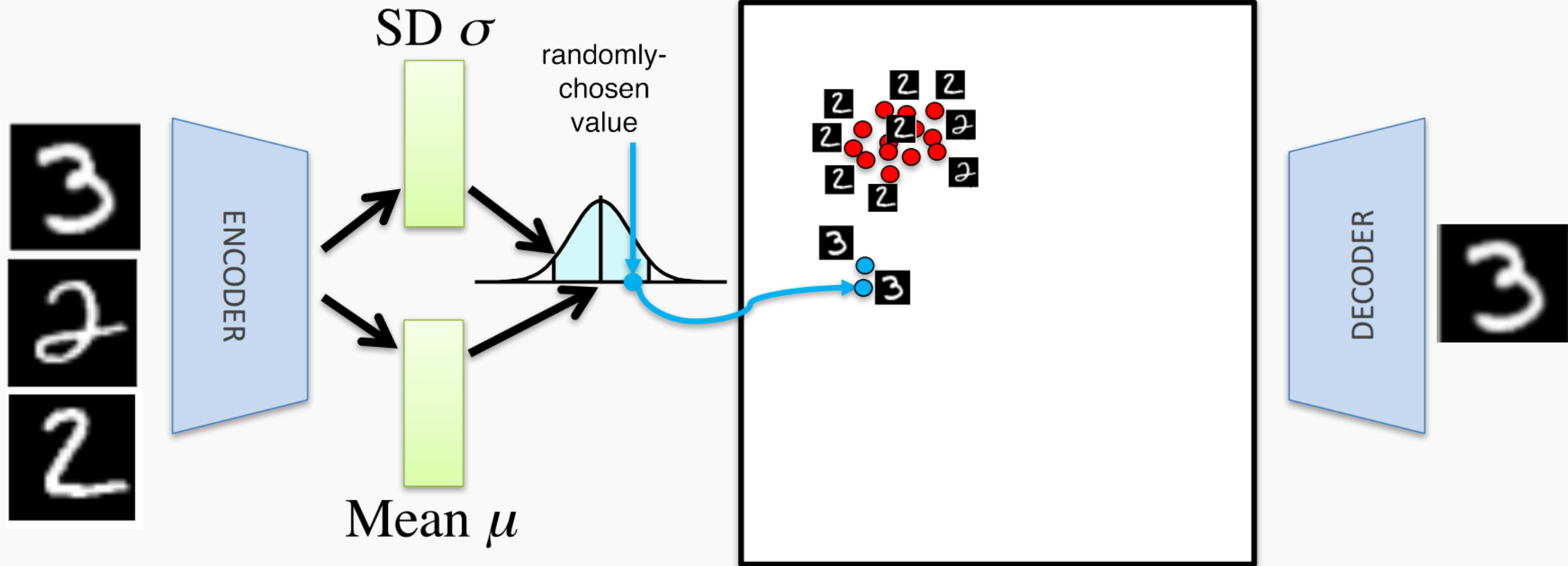
Now lets test again



Variational Autoencoder

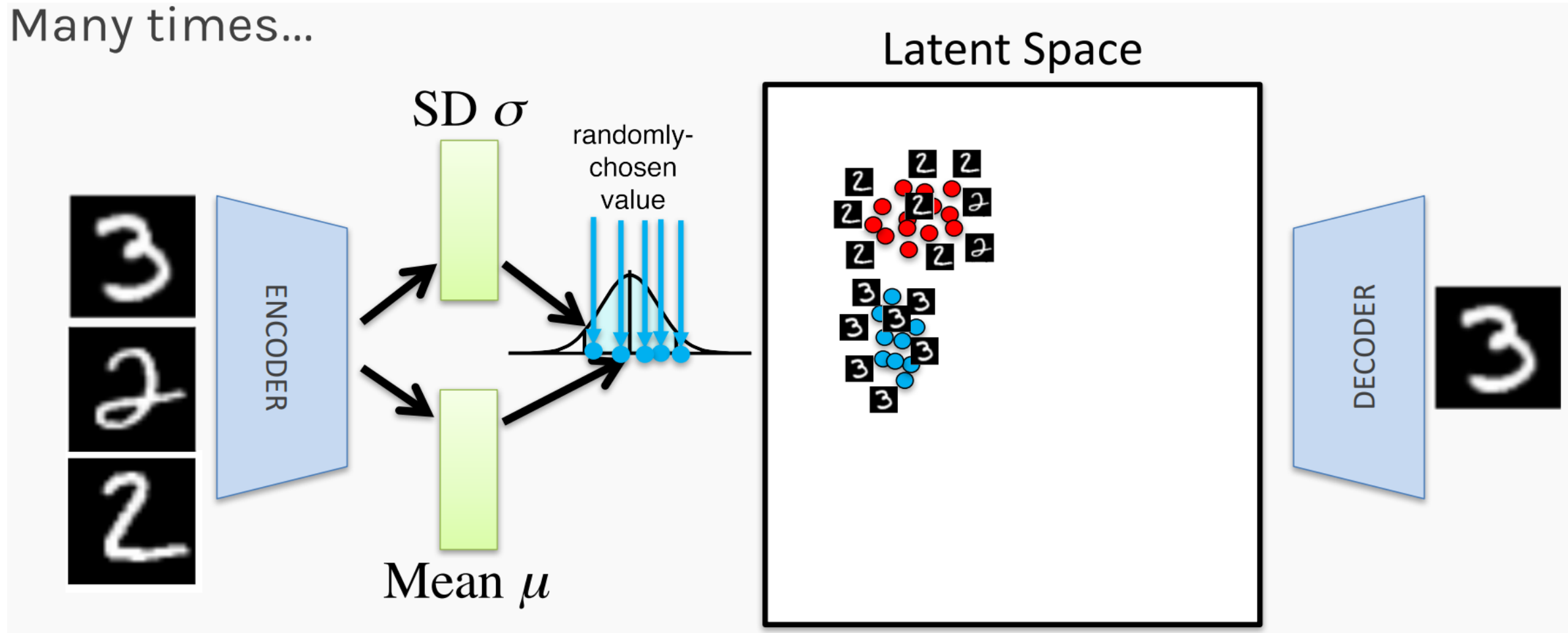
Training on 3's again

Latent Space

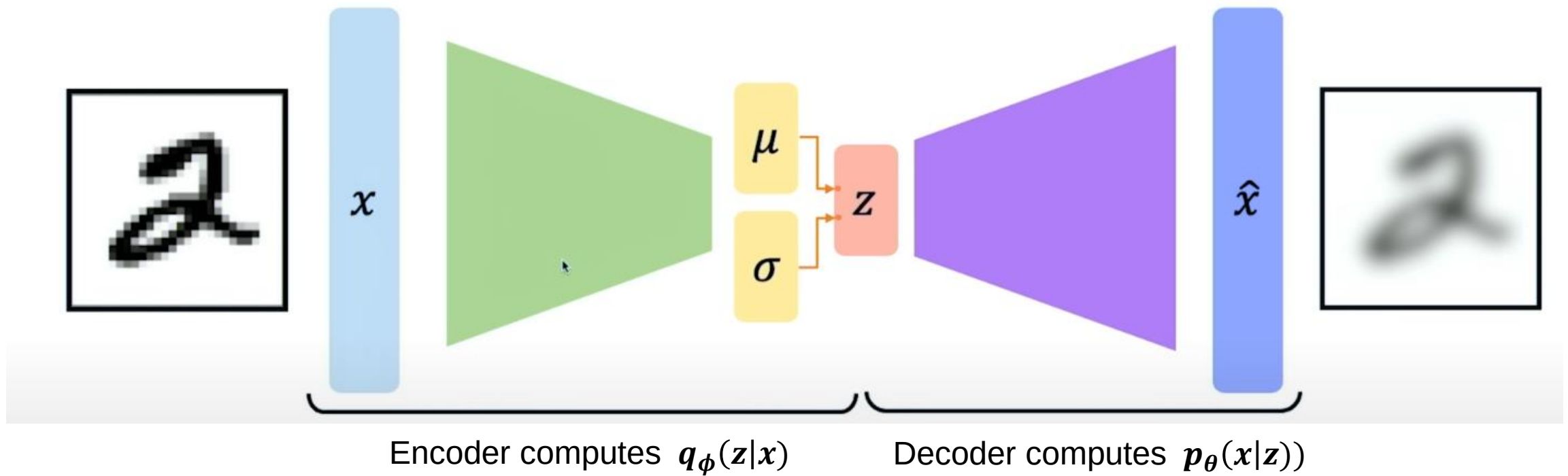


Variational Autoencoder

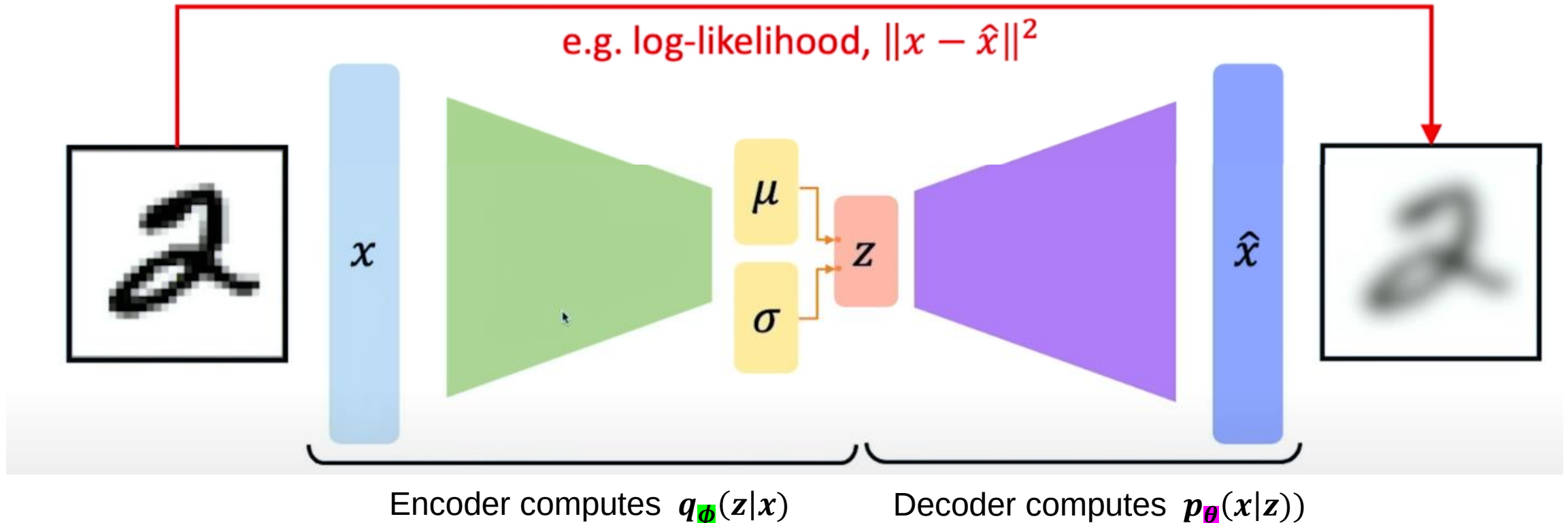
Many times...



Variational Autoencoder Optimization

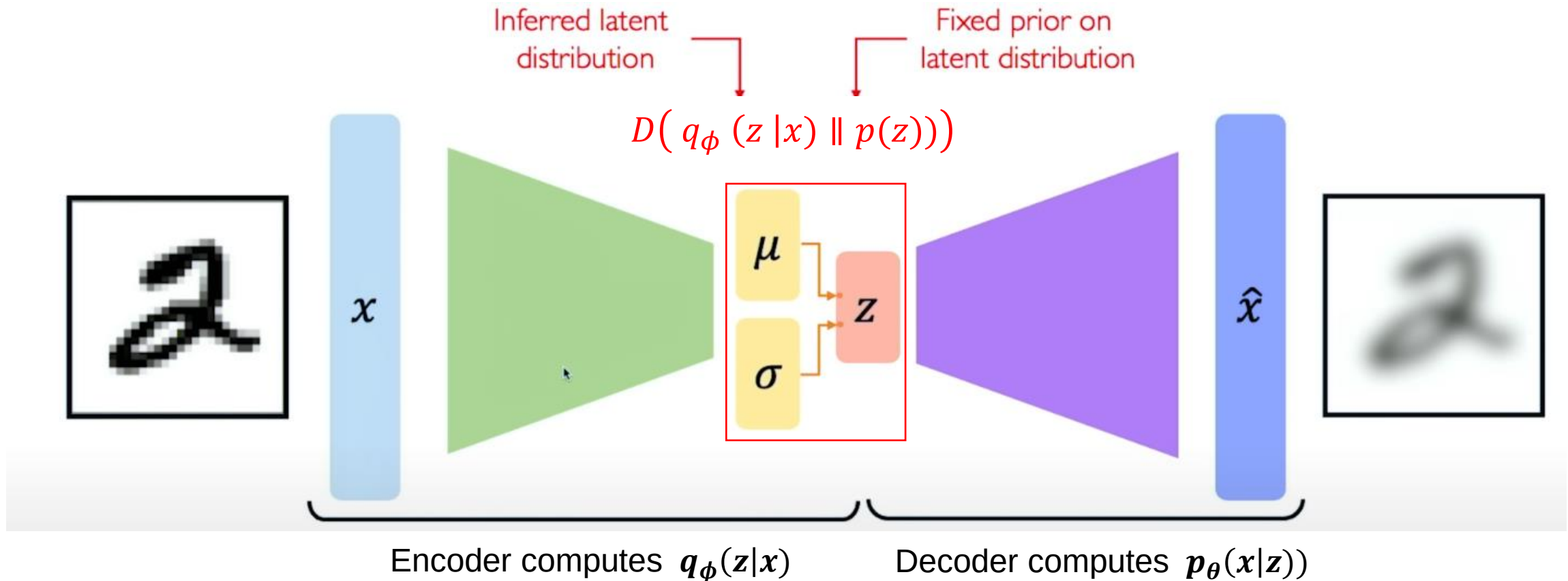


Variational Autoencoder Optimization



- $\mathcal{L}(\phi, \theta, x) = \text{Reconstruction loss} + \text{Regularisation term}$

Variational Autoencoder Optimization



- $\mathcal{L}(\phi, \theta, x) = \text{Reconstruction loss} + \text{Regularisation term}$

Variational Autoencoder

Optimization – Prior on latent distribution

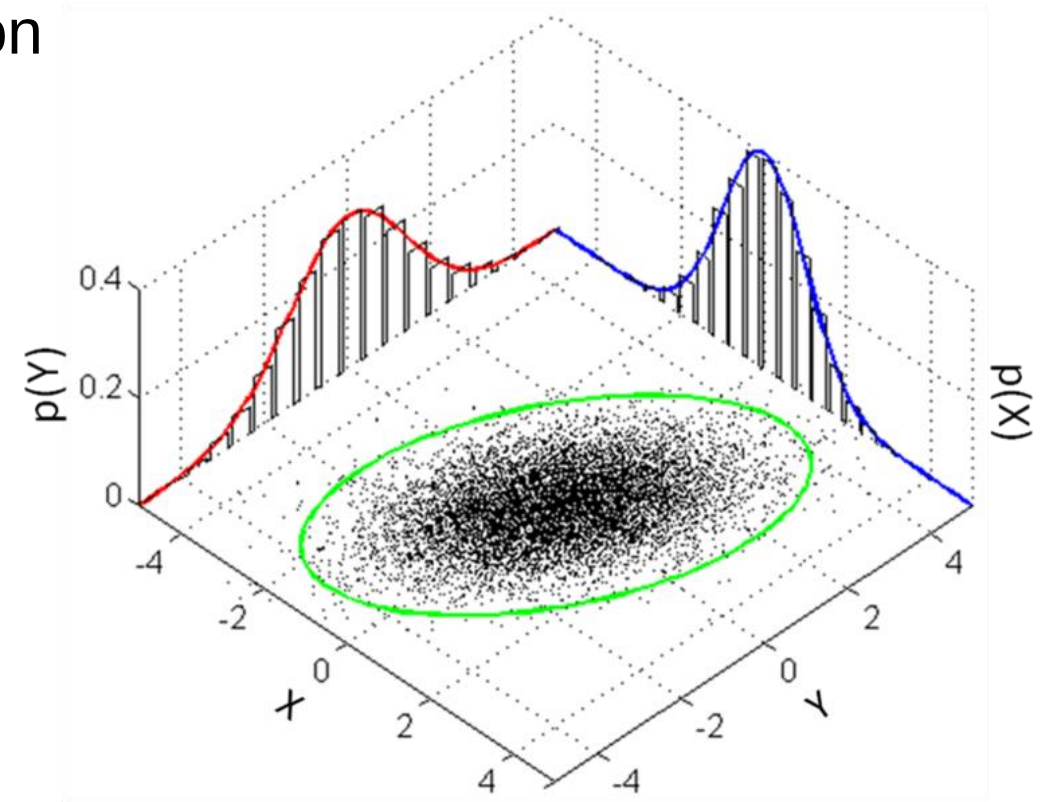
- Common choice of prior – Normal distribution

$$p(z) = \mathcal{N}(\mu = 0, \sigma^2 = 1)$$

- Encourages encoding to distribute encodings evenly around the center of the latent space
- Penalize the network when it tries to cheat by clustering points in specific region (i.e. By memorizing the data)

- KL Divergence

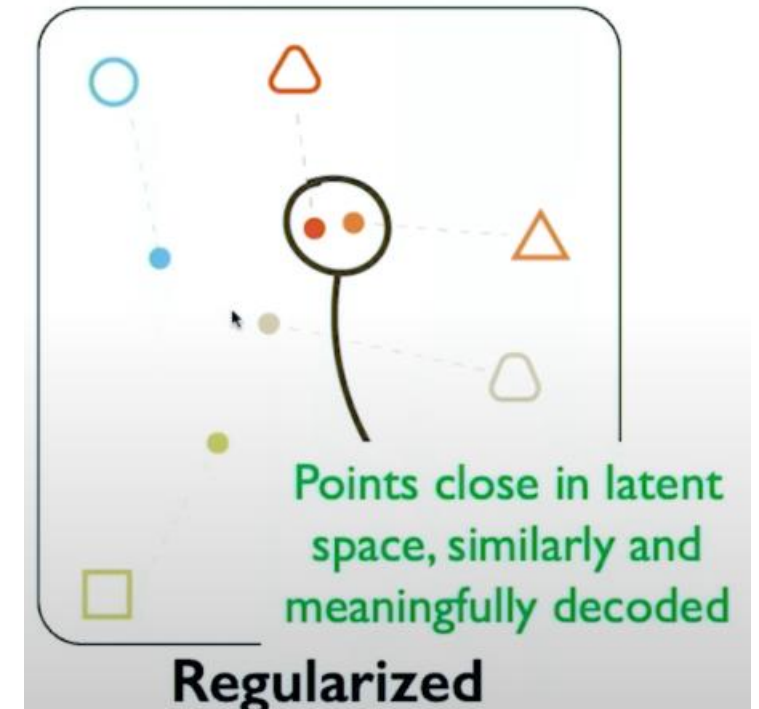
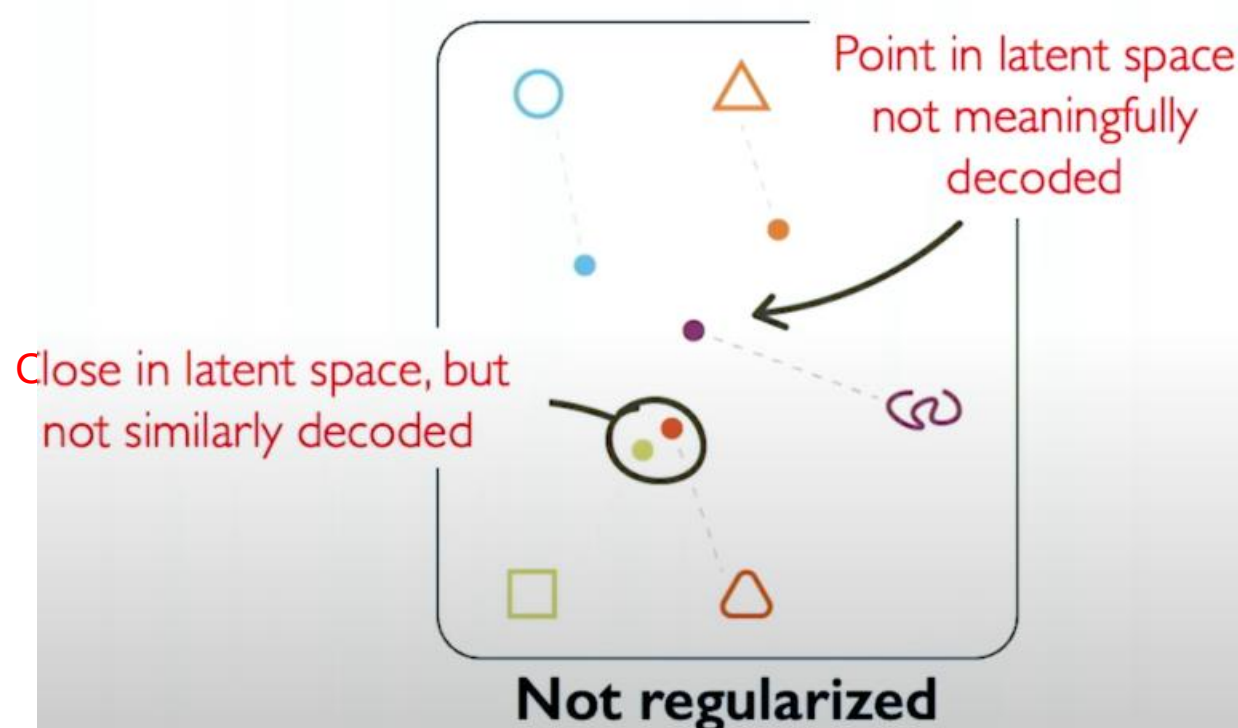
$$D(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_{j=0}^{k-1} (\sigma_j^2 + \mu_j^2 - 1 - \log \sigma_j)$$



Variational Autoencoder

Optimization – Intuition on regularization and normal prior

- Regularization
 - Continuity: Points that are close in latent space $\rightarrow$ Similar content after decoding
 - Completeness: Sampling from latent space $\rightarrow$ “Meaningful” content after decoding



Variational Autoencoder

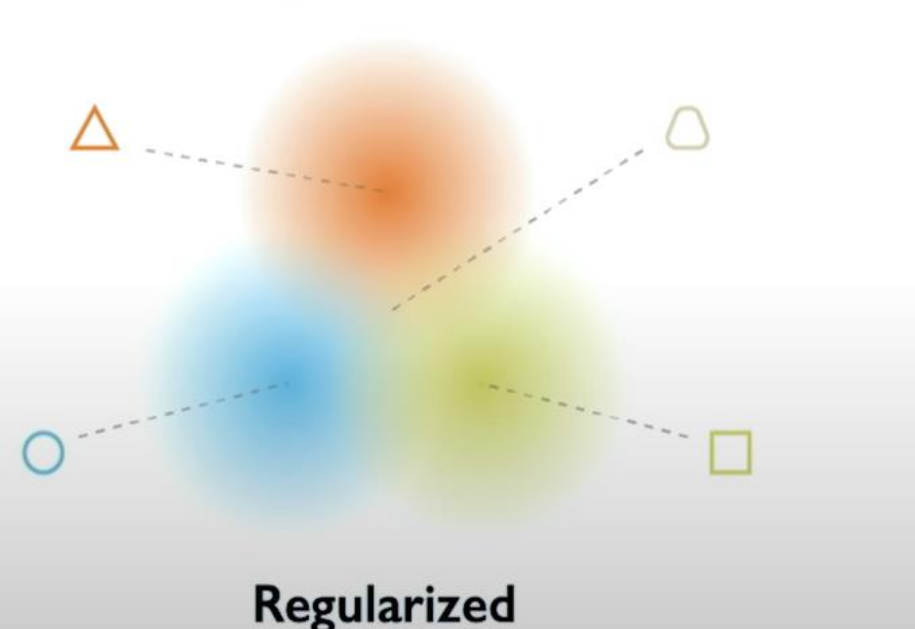
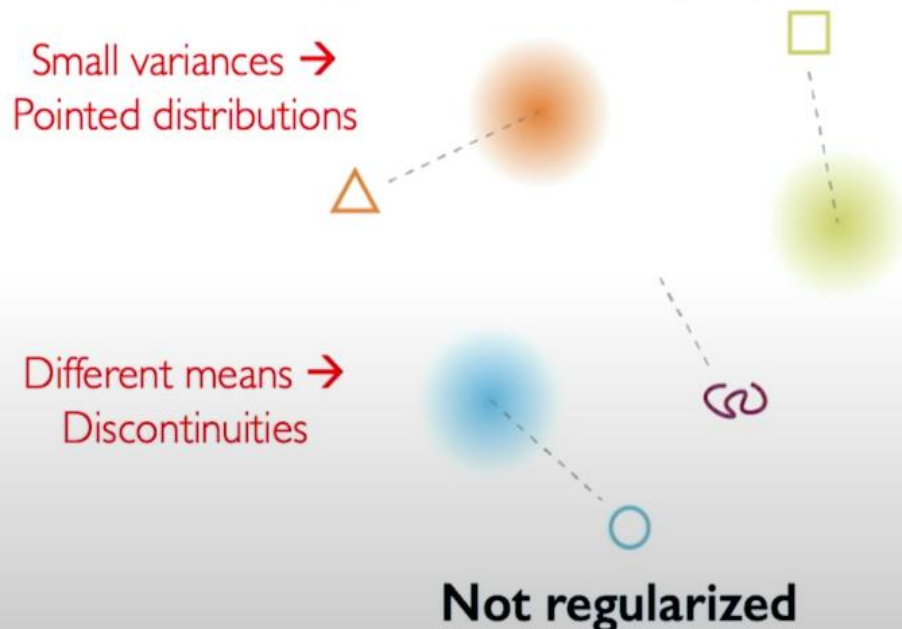
Optimization – Intuition on regularization and normal prior

■ Regularization

- Continuity: Points that are close in latent space $\rightarrow$ Similar content after decoding
- Completeness: Sampling from latent space $\rightarrow$ “Meaningful” content after decoding

Encoding as a distribution does not guarantee these properties!

Normal prior $\rightarrow$ continuity + completeness



Variational Autoencoder

Optimization – Intuition on regularization and normal prior

■ Regularization

- Continuity: Points that are close in latent space $\rightarrow$ Similar content after decoding
- Completeness: Sampling from latent space $\rightarrow$ “Meaningful” content after decoding



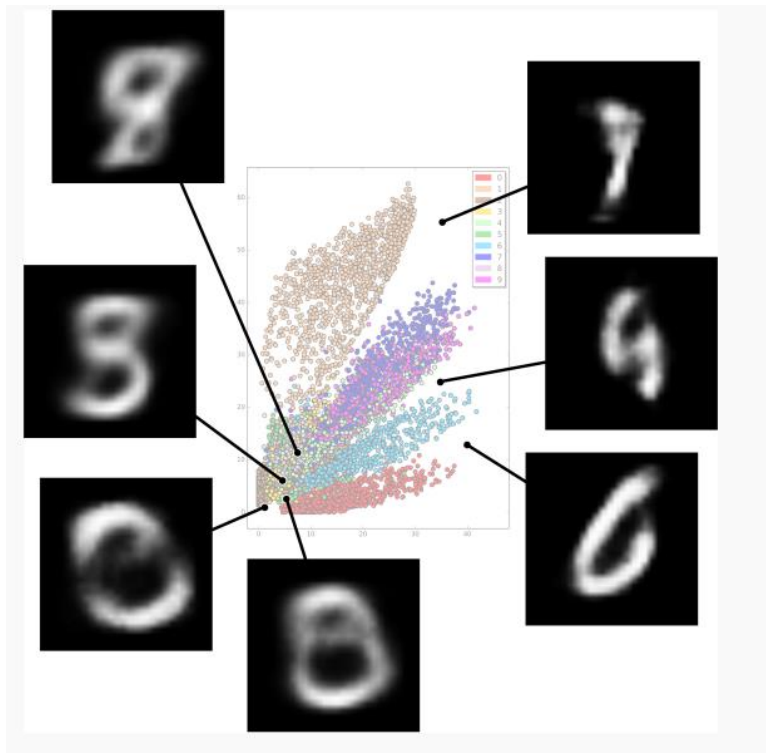
Regularization with normal prior helps enforce information gradient in the latent space!

Variational Autoencoder

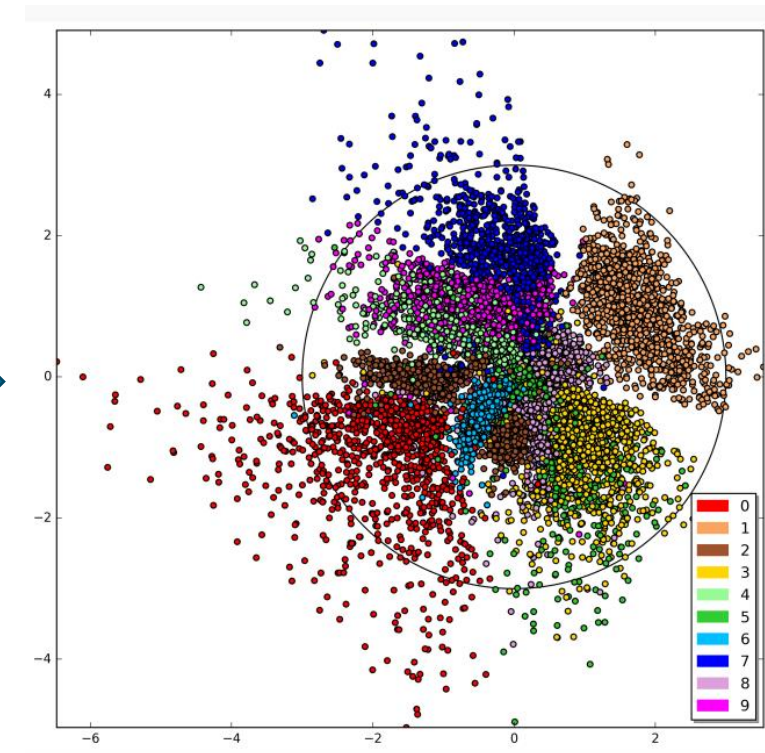
Optimization – Intuition on regularization and normal prior

■ Regularization

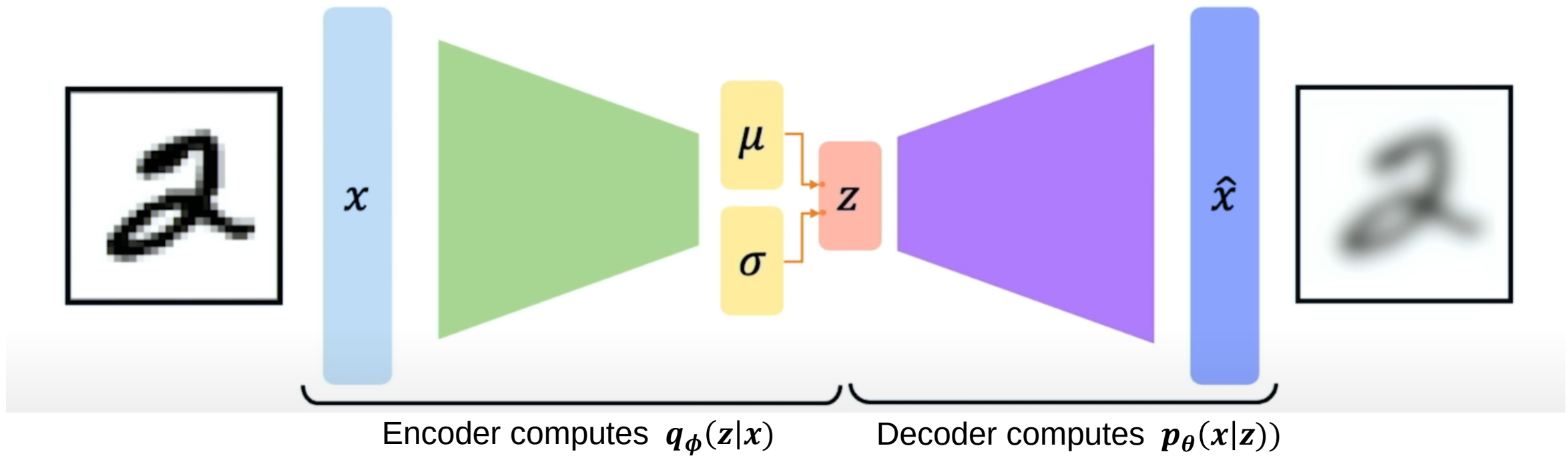
- Continuity: Points that are close in latent space $\rightarrow$ Similar content after decoding
- Completeness: Sampling from latent space $\rightarrow$ “Meaningful” content after decoding



Regularization

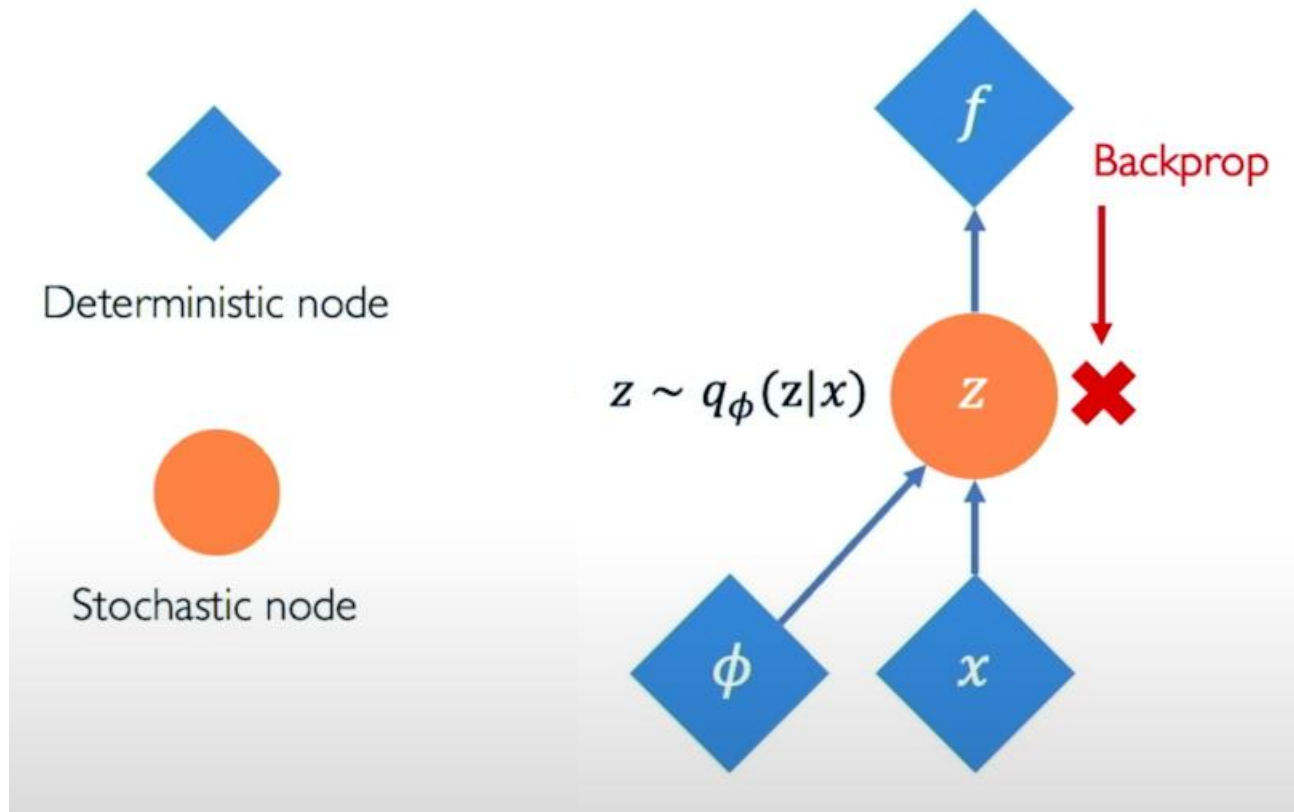


Variational Autoencoder Optimization



Variational Autoencoder

Optimization – Reparametrizing the sampling layer

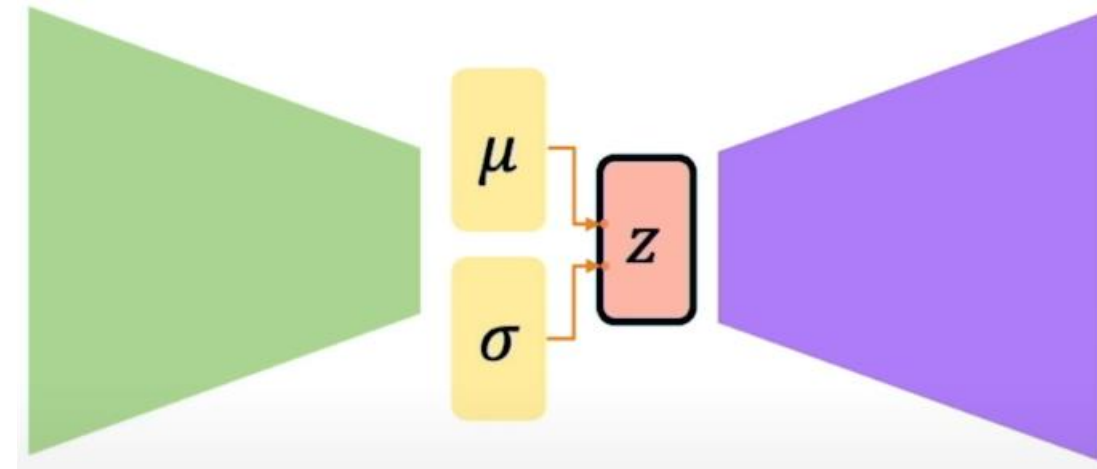


Variational Autoencoder

Optimization – Reparametrizing the sampling layer

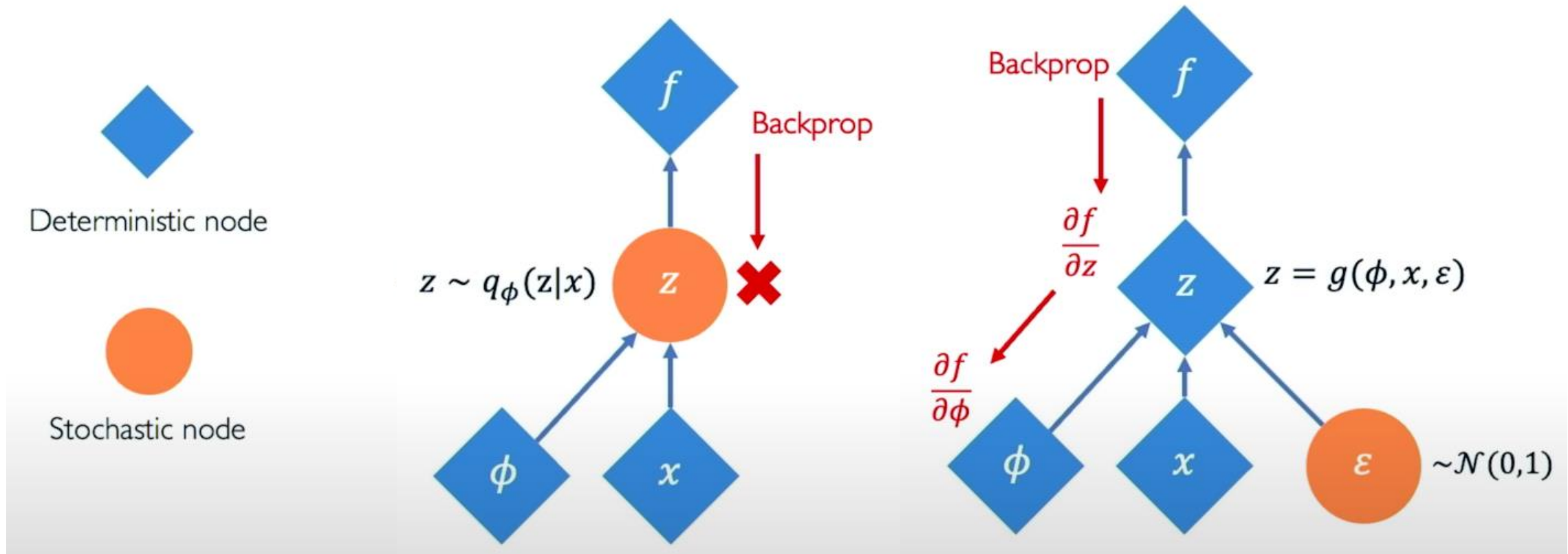
- Sampled latent vector z
 $z \sim \mathcal{N}(\mu, \sigma^2)$
- Consider z as the sum of
 - A fixed μ vector
 - And fixed σ vector, scaled by random constant drawn from the prior distribution

$$z = \mu + \sigma \odot \epsilon$$



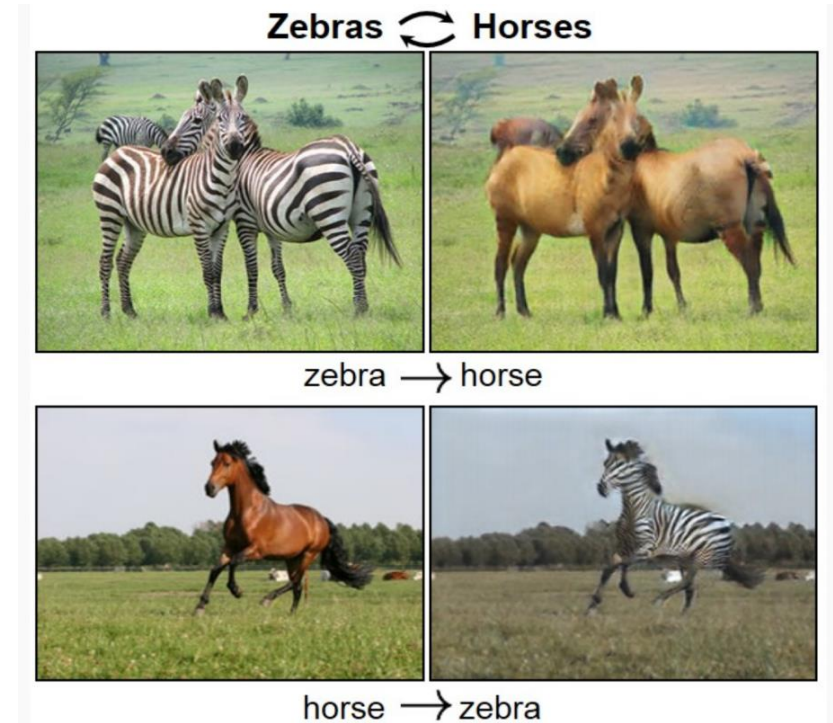
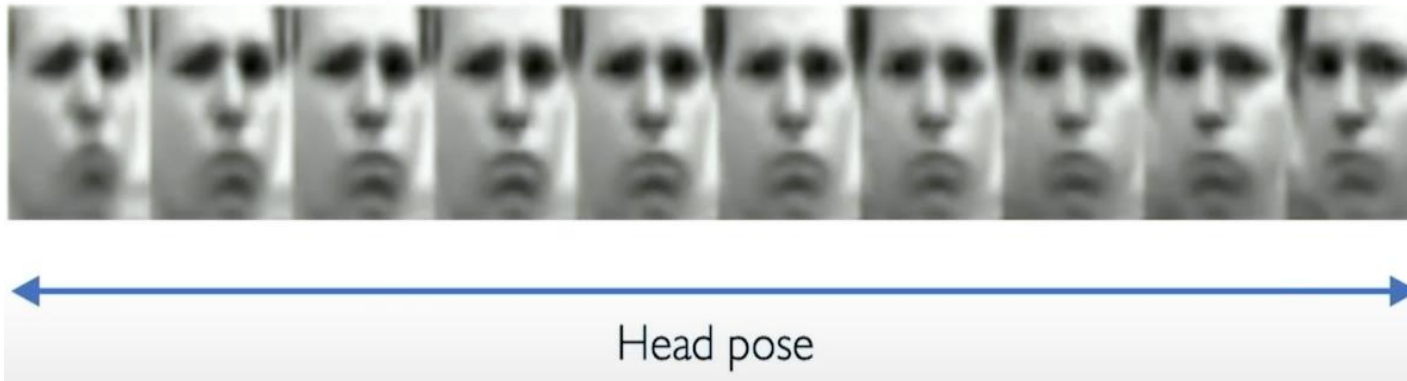
Variational Autoencoder

Optimization – Reparametrizing the sampling layer



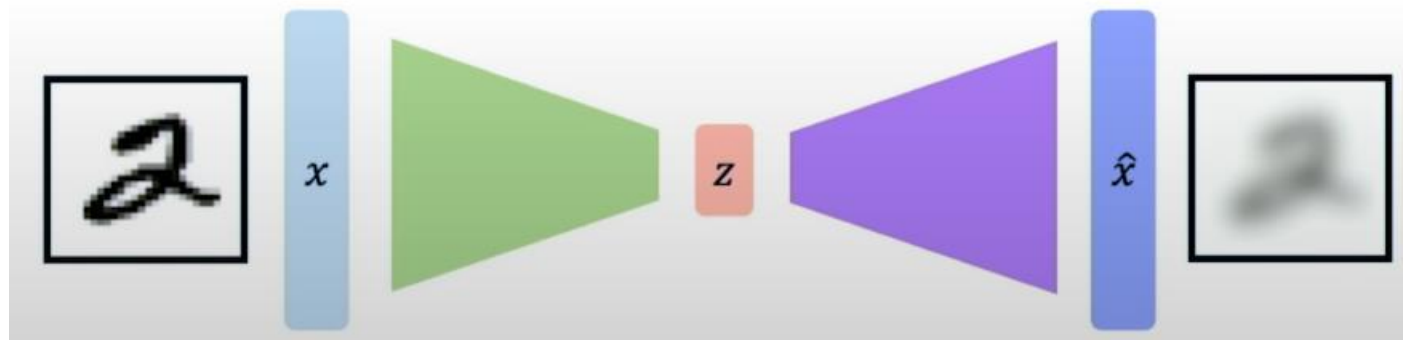
Variational Autoencoder Optimization – Latent perturbation

- Different dimensions of z encodes different interpretable latent features
- Slowly increase or decrease a single latent variable while keep all other variables fixed



Variational Autoencoder Summary

- Compress the representation of world to something we can use to learn
- Reconstruction allows for unsupervised learning (no labels!)
- Reparameterization trick to train end-to-end.
- Interpret hidden latent variables using perturbation
- Generating new samples.



Practical Applications

1. Satellite Monitoring (TerraSAR-X / TanDEM-X)



Monitor satellite telemetry and
detect anomalies in real time



Examples: Power subsystem
degradation, thermal anomalies,
communication faults

2. UAV Flight Monitoring (superArtis)



Detect abnormal flight behavior
and system faults



Examples: Unusual maneuvers,
sensor failures, actuator
degradation

Detecting irregularities – The CausalAnomalies project

Practical Applications

3. Infrastructure Monitoring (Railway Systems)



Identify anomalies in railway infrastructure and operations



Examples: Track irregularities, switch failures, signal malfunctions

4. IT & Cybersecurity Log Monitoring



Detect malicious activities and system anomalies in logs



Examples: Intrusion attempts, abnormal access patterns, system misuse

5. Aircraft System Health Monitoring



Monitor aircraft systems and predict potential failures



Examples: Engine degradation, hydraulic faults, avionics malfunctions

Detecting irregularities – The CausalAnomalies project

Next:

- Hands-on
- Theory Part IV: Latest application, code and knowledge sources

Topic: **Introduction to Deep Learning**
Part III – Deep Generative Models

Date: 2026-05-29

Author: Auliya Fitri, Sheetal Sahoo, Sreerag Naveenachandran

Institute: Data Science

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